R2C-MGR2L: Enhancing Multigranularity Relational Representation in Dual-Temporal Optical Remote Sensing Images Through LLM-Based Scene Graph Generation

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***Abstract*—Recent advances in Earth observations have enabled the acquisition of high-precision optical remote sensing imagery (ORSI) with submeter spatial resolution and hourly revisit periods. While early neural network-based methods have demonstrated notable limitations in visual perception, more recent multimodal fusion approaches still struggle to effectively recognize and reason over geospatial object relations. To fully promote the collaborative analysis of very-high-resolution (VHR) dual-temporal ORSIs, this article presents a novel framework named R2C-MGR2L that is designed to enhance multigranularity relational representation in dual-temporal optical remote sensing (ORS) images through LLM-based scene graph generation (SGG). This framework presents a two-stage learning paradigm—knowledge alignment and cross-modal fusion—that incorporates structured knowledge representations and a hierarchical attention mechanism to improve the refinement and reasoning of fine-grained relational semantics. First, we investigate the types of geospatial object relations and assess the feasibility of enhancing relational semantic representation through multimodal knowledge extraction and embedding. We subsequently employ a pretrained code large language model (Code-LLM) to transform multigranularity relation triplets from extracted ORS scene graphs into semantically equivalent programs, thereby alleviating the scarcity of domain-specific ORSI knowledge. Moreover, a hybrid-structured asymmetric contrastive learning network is developed to further refine the semantic representations derived from complex scenes. By hierarchically fusing dual-temporal ORSIs and corresponding scene graphs, an integrated representation of visual targets and structural knowledge across multiple depths is achieved. We evaluate our proposed framework on three relation-aware ORS scene understanding tasks. The experimental results suggest that the R2C-MGR2L outperforms the state-of-the-art generic methods, resulting in consistent improvements in accuracy, completeness, and timeliness for relational semantic recognition.**

***Index Terms*—Dual-temporal optical remote sensing (ORS) image, hybrid-structured asymmetric contrastive learning network, relational semantic representation, scene graph generation (SGG), scene understanding.**

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I. Introduction

THE rapid development of Earth observation technologies, such as optical satellites, aerial platforms, and ground-based sensors, has enabled the real-time acquisition and comprehensive utilization of vast amounts of multisource optical remote sensing imagery (ORSI). As a crucial component of ORSI applications, scene understanding aims to identify diverse geospatial objects in ORSI and analyze their spatiotemporal and semantic associations, thereby marking a transition from feature extraction to scene cognition. Early studies focused primarily on visual feature extraction methods for geographic entities, such as convolutional neural networks (CNNs) [1], which establish mappings between visual features and category labels to support perception tasks including scene classification [2], [3], [4], object detection [5], [6], [7], and semantic segmentation [8], [9]. However, visual representation suffers from significant perceptual limitations: the functional, structural, and ecological characteristics inherent to geospatial objects cannot be fully captured through visual modalities alone, and category labels are inherently inadequate for comprehensive semantic understanding. Although current pretrained cross-modal models [10], [11], [12] provide enhanced semantic expression by fusing multiple ORSI sources, they exhibit notable weaknesses in terms of multiscale relation recognition and complex semantic reasoning. For example, EarthGPT [13] achieves multisensor image comprehension through visual-textual alignment but lacks explicit relational structure modeling among geospatial objects.

Unlike the above modal-based representation learning methods, structured semantic modeling presents a systematic integration of semantic relations among various elements within optical remote sensing (ORS) scenes through well-defined structures such as syntax trees, semantic networks, and logical relations, further generating comprehensive scene descriptions that encompass both spatial distribution patterns and functional semantic roles. Scene graphs, as a structured modeling approach, represent entities as collections of subject–relation–object triplets, effectively bridging the gap between feature representations and human cognition while demonstrating significant advantages in cognitive tasks such as semantic retrieval [16], [17], image captioning [18], [19], and visual question answering [20], [21]. Recently, large language

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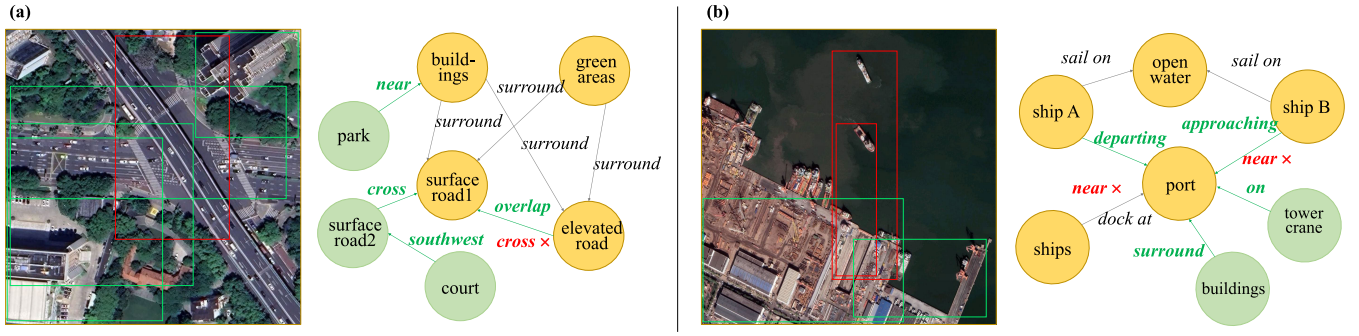


Fig. 1. Common issues in SGG. (a) Incomplete expression and inaccurate modeling of geospatial object relations. (b) Insufficient temporal modeling of geospatial object relations. The ORS scene graphs generated by GPT-4.1-2025-04-14 are highlighted in yellow, while the fine-grained geospatial object relations are highlighted in green.

model (LLM)-based scene graph generation (SGG) methods [7], [22], [23], [24], [25], [26] have typically formulated scene graph construction as a structured program modeling task, automatically extracting subject–relation–object triplets to construct semantically rich scene subgraphs by leveraging the contextual reasoning capabilities of LLMs. For instance, LLM4SGG [7] leverages LLMs for weakly supervised pseudo-label generation from image captions, whereas LLM meets Scene Graph [23] benchmarks the capability of LLMs to understand and generate scene graphs in textual settings and introduces a causal adjustment module for debiasing SGG. Notably, large-size very-high-resolution (VHR) ORSIs often contain abundant multiscale geospatial objects with diverse spatial relations, posing a key challenge for the construction of structured models. Moreover, despite their powerful multimodal reasoning capabilities, current mainstream LLMs are primarily trained on large-scale general-domain corpora and lack essential domain-specific knowledge of ORSIs—including professional classification systems, functional semantic characteristics, and spatiotemporal evolution patterns—which significantly limits their multimodal understanding and scene adaptability. For instance, RS-RAG [28] incorporates domain-specific prior knowledge through retrieval-augmented generation to increase reasoning accuracy; however, delayed updates and coarse-grained semantic representations affect its retrieved knowledge. Consequently, LLM-based SGG remains insufficient for modeling complex relations among multiscale geospatial objects in VHR ORSIs. Specifically, general-purpose LLMs face three core challenges when applied to remote sensing SGG.

1) *Completeness:* In VHR ORSIs, geospatial objects span several orders of magnitude in spatial extent. This extreme scale diversity renders it difficult to simultaneously capture fine-grained small targets, large-scale structures, and the semantic associations among them. As shown in Fig. 1(a), GPT-4.1-2025-04-14—one of the most capable LLMs currently available—identifies relations among buildings, roads, and green areas but fails to discover relations between the court and the road, or between the park and the building.
2) *Accurateness:* Current mainstream LLMs lack critical domain-specific knowledge of ORSIs—including professional classification systems, functional semantic characteristics, and spatiotemporal evolution patterns—limiting them to handling generic spatial relations and preventing the capture of deeper semantics such as functional attributes, behavioral intentions, and environmental contexts. As illustrated in Fig. 1(a), GPT-4.1-2025-04-14 misclassifies an elevated highway as a ground-level road, producing a spurious intersection relation.
3) *Timeliness:* Most existing methods perform frame-by-frame inference on single-temporal images, capturing only static spatial states while neglecting temporal correlations. However, dynamic relations among geospatial objects in ORSIs are often implicit and difficult to perceive directly. As a result, these methods cannot effectively model temporal dynamics such as motion trajectories, state evolution, and interactive behaviors. As demonstrated in Fig. 1(b), two ships enter and leave the port, yet GPT-4.1-2025-04-14 captures only their static state near the port and fails to characterize their temporal movement dynamics.

In summary, we argue that the root causes of these issues stem from the dual constraints arising from data annotation and spatiotemporal modeling. First, existing ORSI datasets are restricted in terms of both scale and semantic coverage. Their knowledge annotations lack dynamic behaviors and temporal semantics. Second, mainstream ORS scene understanding relies on single-temporal images for frame-wise inference, neglecting multitemporal feature correlations. Although knowledge graph (KG) enhanced approaches such as RS-RAG [28] and StreamingRAG [29] have improved reasoning accuracy and temporal adaptability by incorporating domain-specific prior knowledge, they remain limited by inherent bottlenecks—including delayed updates, coarse-grained semantic representations, and weak integration with the LLM's reasoning chain—thereby constraining their capacity for joint modeling of complex spatial configurations and dynamic semantic relations. Inspired by neurosymbolic learning, we propose a two-stage learning framework (R2C-MGR2L) that facilitates multigranularity relational representation in dual-temporal ORS images through LLM-based SGG. The first stage focuses on recognizing and structuring

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multiscale geospatial object relations in ORSIs. A model with an identical architecture independently processes temporally adjacent images from the image sequence, generating corresponding scene graphs for each temporal phase. By leveraging the in-context learning and physical world modeling capabilities of code large language models (Code-LLMs), various geospatial object relations are converted into semantically equivalent executable programs that enable standardized knowledge encoding. The obtained scene graphs encode potential causal relations and are subsequently fed into the second stage, where a dual-encoder network is designed based on the constructed scene graph knowledge for joint representation of dual-temporal ORSIs and structured texts. Through hierarchical feature alignment, the representation accuracy for multiscale geospatial object relations in complex ORS scenes is substantially improved. This two-stage design enables the model to focus more effectively on capturing the changes reflected in dual-temporal ORSIs rather than being confounded by the complexity of simultaneous multitemporal processing. The main contributions of this article are summarized as follows.

1) We propose R2C-MGR2L, an innovative two-stage framework that enables multigranularity relational representation in dual-temporal ORS images through LLM-based SGG.
2) To enhance external knowledge in ORSI, we present a novel structured knowledge generation method, which transforms the integration of multisource heterogeneous ORS knowledge into a structured knowledge generation task by leveraging a Code-LLM. Furthermore, to effectively represent evolutionary multiscale relations in dual-temporal ORSIs, we present an asymmetric contrastive learning network with a hybrid architecture (GEO-H2ACLNet), which employs hierarchical fusion for the joint representation of visual geospatial objects and structured knowledge across different feature depths.
3) We introduce the first dual-temporal image–text dataset (DTITD130H) designed for fine-grained geospatial object relational semantics understanding in dual-temporal ORSI. Moreover, extensive large-scale experiments are conducted to validate the superiority of our framework on three relation-aware downstream tasks. The results show that compared with generic state-of-the-art approaches, the R2C-MGR2L achieves superior performance in terms of completeness, accuracy, and timeliness of relational semantic recognition.

II. Related Work

In this section, we review the current literature from three aspects: RS scene relation understanding, RS scene representation, and knowledge generation. These three lines of research collectively define the technical landscape for advancing RS scene understanding from isolated object recognition toward structured, semantics-rich relation modeling. We analyze the progress and limitations of each aspect to motivate the design of our approach.

A. RS Relation Understanding

The traditional relation understanding methods evolved from using low-level visual features such as texture, color, and shape [30], [31], [32] to applying deep learning techniques for extracting high-level features [33], [34], [35], [36], [37], [38]. However, these methods focus mainly on the independent features of geospatial objects while neglecting their spatial dependencies and semantic associations. Early approaches modeled RS geospatial object relations using geometric rules, including distances, orientations, and geometric contacts [39]. Later studies mainly adopted graph neural networks (GNNs) to model positions, topological structures, and co-occurrence patterns [40], [41]. Although GNNs provide strong structured abstractions for explicit spatial relations, their ability to model complex higher order semantics remains limited. Inherent constraints in graph structures hinder the integration of multimodal semantic priors with dynamic logical reasoning processes, leading to inadequate representation of functional associations and spatiotemporal dependencies. Scene graphs provide fine-grained, interpretable relation understanding frameworks by representing semantic associations through triplets. Ren et al. [42] proposed a scene graph similarity-based RS image retrieval algorithm that employs a neural tensor network for similarity calculation of graph feature vectors. Wang et al. [16] developed a scene graph-aware network that integrates multimodal features from images, graphs, and text for retrieval. Tang et al. [17] introduced a spatial relation model using geographic knowledge, improving retrieval accuracy. Recently, the rapid development of LLMs has led to the emergence of a new pathway [43]. Yang et al. [44] migrated LLMs and VLMs to convert scene graphs into natural language captions for RS image retrieval.

B. RS Scene Representation

RS scene representation is aimed at mapping the results of scene relation understanding, which includes geospatial objects, spatial structures, and semantic relations, to a unified feature space. Recently, multimodal vision-language approaches for RS scene representation have become popular research topics. PIR-CLIP [46] is a prior knowledge-guided image–text representation mechanism that enables the dynamic filtering and enhancement of semantic features. RemoteCLIP [47] learns semantically rich visual features and aligned embeddings, demonstrating strong cross-task transferability. MG-Transformer [48] uses grouped attention to fuse CLIP image embeddings, addressing the scarcity of image–text pairs. RSGPT [49] efficiently aligns visual and textual features by conducting lightweight fine-tuning on its Q-Former module and linear layers. RS-LLaVA [50] adapts LLaVA to RS images using a low-rank adaptation, significantly improving the performance achieved in RS image captioning and RS visual question answering. Aquila [51] introduced a multiscale spatial feature fusion mechanism to capture multiscale information, and a hierarchical alignment strategy to achieve deep feature alignment. SkySenseGPT [52] adopts region-level feature encoding and multimodal instruction alignment, reinforcing performance its detailed description and multitask

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TABLE I
SUMMARY OF LITERATURE REVIEW

Work	Design objective	Methodology	Advantages	Disadvantages
[39]	High-resolution RS scene classification	Semantic-aware graph network	Semantic exploitation and unstructured relation mining	Limited performance under high semantic similarity
[40]	Non-semantic interference elimination in RS change detection	Coordinate space and feature interaction-graph convolutional network	Accuracy-efficiency balance with nonsemantic variation identification	Limited unsupervised / weakly supervised capability
[41]	Global context and edge feature extraction for RS change detection	Attention-aware Sobel GCN	Enhanced edge feature extraction	Labeled data dependency and limited generalization
[16]	Image consistency and difference modeling	Scene graph-aware hierarchical fusion	Multimodal feature integration for retrieval	Inefficient for large-scale feature maps
[17]	Semantic alignment under sparse image-text pairs	Language-empowered conversion pipeline	Structured-to-unstructured modality gap bridging	Unexplored optimal prompt strategies
[45]	RS vision-language representation for open-domain retrieval	Two-stage prior instruction representation learning	Scene recognition prior knowledge leverage	Limited long-term dependency modeling
[46]	Semantic granularity enhancement	Multiscale grouping transformer with CLIP latents	Multiscale and pre-aligned image-text integration	Limited scenario adaptability and high computational cost
[49]	Joint RS image captioning and visual question answering	LLaVA-based multitask framework	Multitask capability for captioning and VQA	Limited counting problem handling
[52]	Zero-shot fine-grained image understanding	Fine-grained vision-language foundation model	Enhanced semantic alignment via fine-grained annotations	Incomplete long-text semantic relation capture
Our work	Multiscale geospatial object relation recognition and reasoning in complex ORS scenes	multi-granularity relational representation in dual-temporal ORSIs through LLM-based SGG	Structured knowledge integration and hierarchical cross-modal fusion	Lack of multimodal distribution balance consideration

generalization performance. LRSCLIP [53] is a long-text-enhanced method based on CLIP that incorporates fine-grained annotations to enhance the degree of semantic alignment between visual and textual features.

C. Knowledge Generation

As the understanding of RS scenes has advanced toward deeper semantics and more complex relation reasoning, Code-LLMs such as Codex [54] and CodeT5 [55] have demonstrated unique advantages in structured knowledge generation tasks. The conversion of natural language into executable code enables Code-LLMs to provide structured knowledge support for geospatial object recognition, scene parsing, and relation reasoning processes. Existing studies have redefined information extraction tasks as code generation problems, including named entity recognition, relation extraction, and event extraction. Li et al. [56] introduced CodeIE, which achieves high accuracy, few-shot extraction through prompt learning with the Codex. Madaan et al. [57] modeled event structure prediction as a form of code generation, enabling efficient event element extraction using Code-LLMs. Bi et al. [58] presented CoCoGEN, which generates highly structured knowledge without fine-tuning, surpassing traditional models. These advances validate the potential of Code-LLMs to perform RS structured modeling. Recently, researchers have investigated the deep integration of KGs with Code-LLMs to improve the efficiency of structured knowledge generation and reasoning schemes. One class of methods employs Code-LLMs to generate rule templates and logical relations, which are then mapped back to KGs for knowledge expansion and structural completion. Agarwal et al. [59] transformed triplet generation into code completion using Code-LLMs with structure-aware prompts and reasoning enhancements. Saberi and Fard [60] reformulated logical form generation as code generation, using modular functions and question-related subgraphs for knowledge awareness. This approach improved the KG understanding capabilities and question-answering accuracy of the model. Another class of methods focuses on KG-driven retrieval-augmented mechanisms, guiding Code-LLMs to generate task-specific code by retrieving contextually relevant triplets. Athale and Vaddina [61] designed a context-augmented framework using programming KGs with semantic retrieval and reranking for improved accuracy. He et al. [62] represented code repositories as KGs to capture structural relations, enhancing Code-LLM context awareness.

To highlight the unique contributions of our approach compared with existing works, we provide a concise comparative overview in Table I, which illustrates key distinctions in terms of design objectives, methodologies, advantages, and disadvantages.

III. METHODOLOGY

In this section, we present the main components of the proposed R2C-MGR2L framework, including ORS relation analysis, ORS scene graph construction, and ORS scene representation. Finally, we describe the multimodal data sampling strategy adopted in this framework.

A. ORS Relation Analysis

The analysis of geospatial object relations in ORS scenes provides a critical foundation for performing geospatial object recognition, semantic reasoning, and scene comprehension. This study divides common geospatial object relations into three categories: spatial relations, semantic relations, and interactive relations. Spatial relations refer to structural connections among geospatial objects in geographic space, comprising topological relations (e.g., “adjacent,” “contain,” and “intersect”), orientational relations (e.g., north and east), and distance relations. These relations reflect the spatial configurations and structural patterns among geospatial objects. Semantic relations focus on the logical associations among geospatial objects in terms of functions, attributes, and scene

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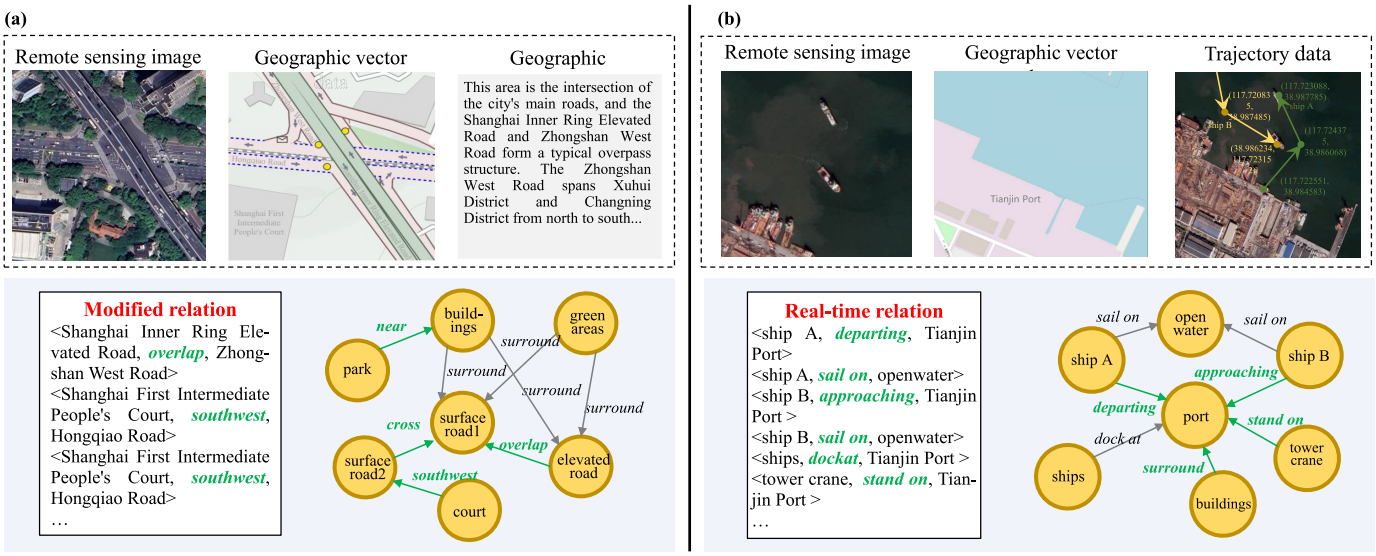


Fig. 2. Illustration of reasoning about fine-grained geospatial object relations based on multimodal knowledge.

context (e.g., "belong to," "connect," and "cover"). These high-level relations reflect the roles and association rules of the geospatial objects contained in specific scenes, which typically rely on human experience for judgment and cannot be directly extracted from ORSIs. Interactive relations reflect the dynamic evolution trends of geospatial objects in the spatiotemporal dimension (e.g., approaching, departing, and passing through). These dynamic relations exhibit distinct temporal characteristics and prove crucial for characterizing the spatiotemporal behavioral patterns of geospatial objects. The incorporation of trajectory data and behavioral analysis logic enables the effective identification and characterization of the interactive relations between moving geospatial objects across different temporal phases. The incorporation of geographic vector knowledge, unstructured geographic texts, and trajectory data enables effective identification of these relations and compensates for ORSI limitations, as shown in Fig. 2. We further propose a unified multimodal remote sensing scene knowledge database (MR2SKD), which integrates multimodal knowledge from geographic vector data, unstructured geographic texts, and trajectory data. The database systematically organizes and stores geospatial object relations. A comprehensive description and structural illustration of MR2SKD can be found in Appendix A.I (see Supplementary Material), available online.

B. ORS SG Construction

We formulate the scene graph construction process as a structured knowledge generation task and transform the multilevel triplets contained in the given scene graph into semantically equivalent programs written in a general-purpose programming language. Temporally adjacent images from the image sequence are independently processed to generate corresponding scene graphs for each temporal phase. The obtained scene graphs encode potential causal relations and are subsequently fed into Stage Two, where a dual-encoder network leverages the constructed scene graph knowledge for joint representation of dual-temporal ORSIs and structured texts. Suppose that we are given a training sample (T, G), where T denotes text that includes an image caption and corresponding knowledge texts, and G denotes the target knowledge structure to be generated. We adopt a pretrained Code-LLM to generate a programmatic graph representation G_E, which is then converted into a scene graph G. Specifically, we define a code-based ontology in a general-purpose language, explicitly declaring multilevel entities and relations. On this basis, we design a prompt template that guides the Code-LLM to learn the definitions of entities, relations, and their internal associations. In addition, we incorporate RAG to dynamically concatenate semantically matched relation knowledge texts and in-context examples, assisting the Code-LLM in generating semantically enriched triplets.

1) Code-Based Ontology Definition: Inspired by event extraction research [56], we encode an ORS scene ontology in Python, formalizing geospatial objects and their relations as executable class structures. We define two abstract base classes: *entity* and *relation*. As shown in Fig. 3(a), the *entity* class represents the entities extracted from texts. The *GeoEntity* class is inherited from *entity* and is used to instantiate both static and moving geospatial objects. Subclasses such as *building*, *natural*, and *transportation* are inherited from *GeoEntity* to instantiate concrete geospatial objects (e.g., buildings, natural areas, and vehicles). These subclasses retain the fundamental attributes of *GeoEntity*. As shown in Fig. 3(b), the *relation* class represents geospatial object relations and is specialized into three subclasses: *SpatialRelation*, *SemanticRelation*, and *InteractRelation*. Each class has multiple subclasses, forming a multilevel inheritance hierarchy. For example, the *TopologicRelation* class is inherited from *SpatialRelation* and instantiates topological relations among geospatial objects. The *Functional Relation* class is inherited from *SemanticRelation* and expresses functional connections among geospatial objects. The *ProximityRelation* class is inherited from *InteractRelation* and captures dynamic,

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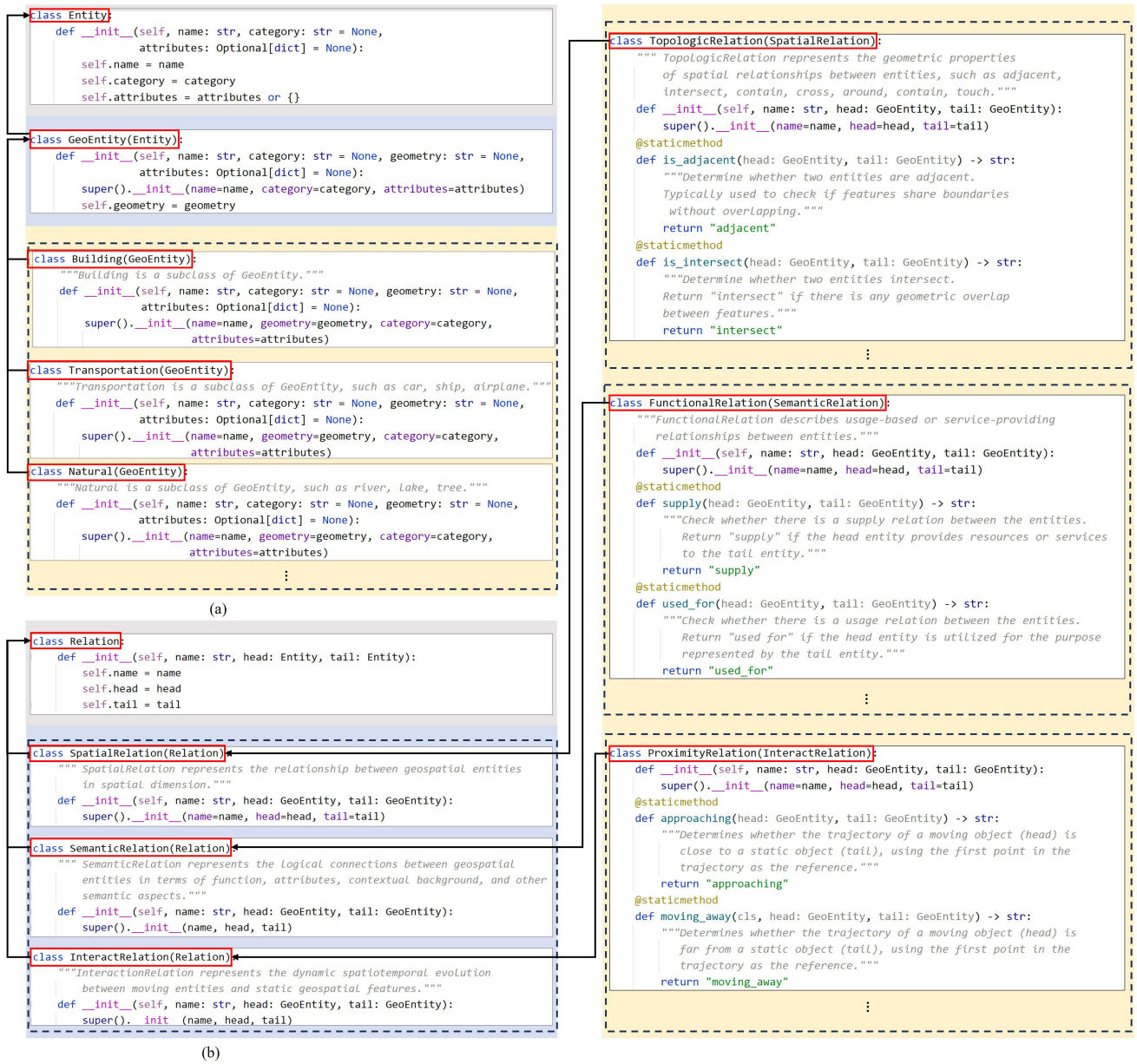


Fig. 3. Ontology definitions of entities and relations. The base class is marked in white, the implementation class in blue, and the subclass in yellow. Arrows indicate inheritance relationships between classes. (a) Entity definition code. (b) Relation definition code.

distance-based interactions between moving geospatial objects and static geospatial objects. Each relation subclass presents static methods that verify whether two geospatial objects satisfy a specific relation, and encapsulates two core attributes, *self.head* and *self.tail*, which form the foundational structure of the triplet ⟨head, relation, tail⟩.

2) ORS SG Generation: We define a prompt template to guide the in-context learning process of the Code-LLM for SGG purposes, as shown in Fig. 4. The API definition employs a unified Python class structure to formalize geospatial objects and multilevel relations. The system prompt provides prior knowledge for reasoning, consisting of retrieved multimodal relation knowledge texts. We employ RAG to supplement the inputs with three types of multimodal relation knowledge texts that are related to the images and image captions: geographic vector texts, trajectory texts, and unstructured geographic knowledge texts. The task instruction describes task requirements in natural language combined with a user query. The correction instruction provides strategies derived from error feedback accumulated during code generation, enabling iterative optimization of the Code-LLM. In-context examples are also retrieved through RAG and serve as few-shot examples to enhance geospatial object relation reasoning and improve the correctness and robustness of the SGG process. More examples of the prompt can be found in Appendix A.II (see Supplementary Material), available online. The context awareness and logical reasoning capabilities of Code-LLM enable the automatic extraction and inference of geospatial object relations in ORS scenes. As illustrated in Fig. 5, the user's query is first embedded and used to retrieve

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API Definition:

[Entity Extraction APIs] [Spatial Relation Reasoning APIs] [Semantic Relation Reasoning APIs] [Interactive Relation Reasoning APIs]

System Prompt:

KNOWLEDGE: These are relation knowledge texts related to the remote sensing image, including Geographic vector data: {Geographic Vector Relation Knowledge Text}, Trajectory Data: {Trajectory Relation Knowledge Text}, and Unstructured Text: {Unstructured Geographic Knowledge Text}. Please infer the relations between geo-objects based on these texts.

Task Instruction:

You are an information extraction expert. List all entities and their relations from the {INPUT}. You must strictly follow the provided API Definition to extract the entities and relations. Each relation type must be associated with its corresponding judgment method as defined in the provided API Definition.

INPUT: This satellite image shows a long, cable-supported road bridge crossing a broad river, and two pylons stand near the banks. On the southeast bank, the bridge ties into a large road junction beside an industrial area with blue-roofed buildings. A small ship is visible in mid-channel beneath the bridge. This image covers the area from (111.384774°E, 30.574096°N) to (111.400901°E, 30.563682°N), and was taken on 2020-02-16T07:30:00.

Correction Instruction:

If the generated triplets are incorrect, please re-identify and infer the relations between entities based on {Error Feedback}.

In-Context Examples:

##Example 1

INPUT: This is a high-resolution satellite view of a cable-stayed bridge crossing a wide river. The bridge deck carries many cars in both directions. On the opposite side of the bridge approach, a green park with circular landscaping and pathways occupies the riverbank. Out on the water, a few ships are underway. This image covers the area from (120.259657°E, 31.954825°N) to (120.271435°E, 31.947246°N), and was taken on 2022-08-10T09:30:00.

OUTPUT:

```
extract = Extract([
  Triplet(
    head=Bridge("Jiangyin bridge"),
    relation=TopologicRelation("crosses"),
    tail=Natural("Yangtze river")
  ),
  Triplet(
    head=Transportation("cars"),
    relation=StativeRelation("running"),
    tail=Bridge("Jiangyin bridge")
  ),
  Triplet(
    head=Leisure("park"),
    relation=OrientationRelation("north"),
    tail=Bridge("Jiangyin bridge")
  ),
  Triplet(
    head=Transportation("ships"),
    relation=StativeRelation("running"),
    tail=Natural("Yangtze river")
  )])
```

Fig. 4. Illustration of the prompt template.

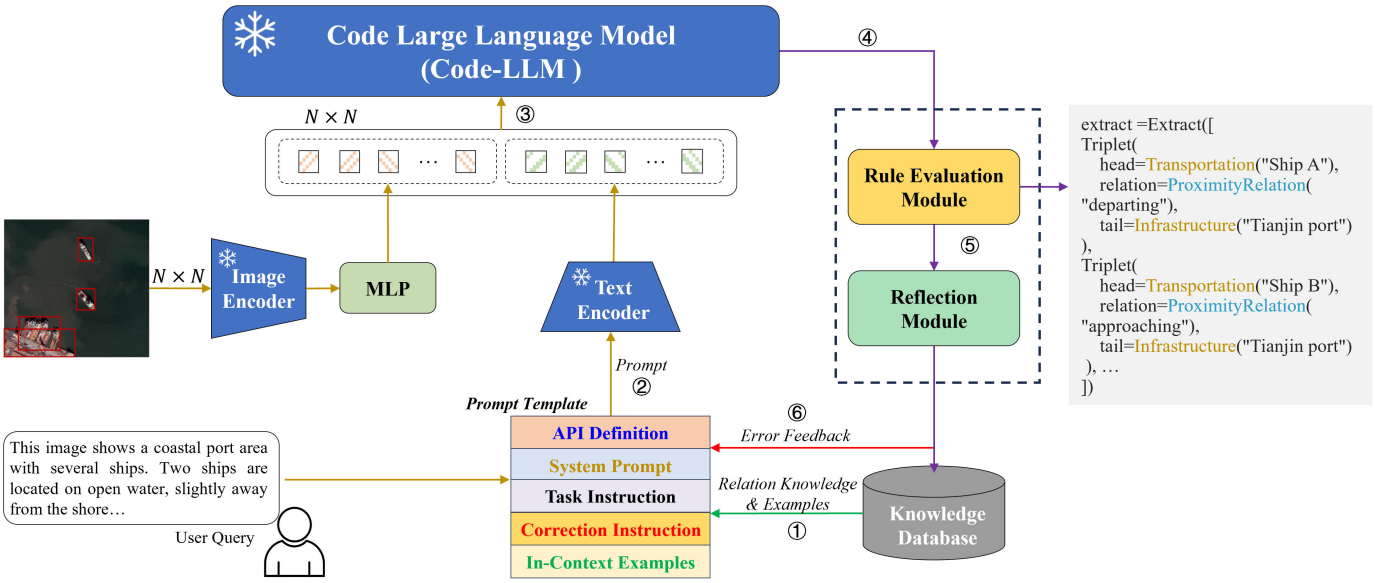


Fig. 5. Process of ORS SGG based on the Code-LLM.

relevant relation knowledge texts and in-context examples via vector similarity search (①). These retrieved items, together with the user's query, are inserted into a prompt template to construct a prompt for conducting in-context learning (②). During the reasoning process, visual tokens and text tokens are concatenated and input into the Code-LLM (③). Through explicit ontology constraints and in-context examples in the prompt, the Code-LLM learns entity classes, relation classes, and their hierarchical relations without parameter updates, thereby enabling relation reasoning and completion (④). The correct triplets generated by the Code-LLM are then converted into a scene graph, in which the geospatial objects form nodes, and the relations at various semantic levels constitute directed edges, producing a scene graph with multigranular geospatial object relations.

We frame the retrieval augmentation process as a search problem. Given a natural language description q, which includes task requirements and an image caption, q is first converted into a vector using an embedding model. An initial candidate set R containing multimodal relation knowledge texts and in-context examples is retrieved from the knowledge database based on vector similarity. To achieve improved

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retrieval quality, we adopt a relevance scoring mechanism that ranks candidates by quantifying the semantic similarity between each candidate and the query. The similarity can be computed as follows:

$$\mathrm{QRsim}(k, q) = \sigma(\mathrm{MLP}(k \parallel q)) \tag{1}$$

where MLP denotes a multilayer perceptron consisting of a fully connected layer and a single-neuron output layer. σ is a sigmoid function, which maps similarity to the interval (0, 1). k is the candidate vector, and $\parallel$ indicates the vector concatenation operation. The top-k most relevant candidates are selected as the final retrieval results.

3) ORS SG Evaluation: To strengthen the reliability of the SGG procedure and suppress hallucinations, we design three types of evaluation rules: consistency rules, authenticity rules, and rationality rules. As shown in Fig. 5, these rules are encapsulated in the rule evaluation (RE) module, which comprehensively validates the generated structured code and assesses the accuracy of the outputs in terms of their structure, factuality, and logic (⑤). These rules are specifically tailored to the structural, factual, and logical characteristics of ORS scenes. The consistency rules verify whether the generated outputs strictly adhere to the code format predefined in the prompt, ensuring that the outputs can be faithfully converted into a scene graph. The consistency rules comprise the relation integrity C_I and the relation inheritance rule C_H, defined as follows:

$$C_I : \forall t \in G_E, t = \langle e_h, r, e_t \rangle \tag{2}$$

$$C_H := \forall r \in R, r \in \mathrm{H}(R_{pre}) \tag{3}$$

where e_h, r, e_t denote the head entity, relation, and tail entity, respectively. Authenticity rules are used to confirm that the expressed geospatial objects and relations have real-world grounding. This verification is particularly critical in ORS, where LLMs may fabricate nonexistent geographic entities or assign spatiotemporally implausible attributes—errors that are difficult to detect through linguistic plausibility alone but can be effectively verified against authoritative geospatial databases. The authenticity rules encompass the entity's existence A_E and the spatiotemporal consistency A_{ST}, defined as follows:

$$A_E : \forall e \in E, e \in M \tag{4}$$

$$A_{ST} : \forall e \in E, ST_Within(\Omega, e) \tag{5}$$

where M denotes the geospatial object collection in the knowledge database, Ω denotes the spatiotemporal range, and ST_Within verifies whether e exists within Ω. Rationality rules focus on domain-specific logic and knowledge constraints to avoid geospatial object relation conflicts that violate domain knowledge. In ORS scenes, geographic entities are governed by strict physical and spatial logic that goes beyond the commonsense reasoning employed in generic SGG and therefore requires explicit rule enforcement. Rationality rules comprise logical rationality and logical orderliness, expressed as follows:

$$R_L : \forall t_i, t_j \in G, t_i \perp t_j \Rightarrow \neg(t_i \wedge t_j) \tag{6}$$

$$R_O : \forall (r_i, r_j) \in R, \tau(r_i) < \tau(r_j) \Rightarrow r_i \prec r_j \tag{7}$$

where $t_i \perp t_j$ indicates mutual exclusivity between t_i and t_j, $\tau(r_i)$ denotes the temporal attribute of r_i, and $r_i \prec r_j$ indicates that r_i occurs earlier than r_j.

We further design a reflective learning mechanism to systematically categorize and analyze the typical errors induced by the Code-LLM, as shown in Fig. 5. A reflection module summarizes these errors, converting them into correction instructions (⑥), incorporated into the prompt—along with the API definition, system prompt, task instruction, and in-context examples—to form a new textual input. This mechanism enables the proactive detection and correction of potential errors during relation reasoning, thereby improving the robustness of the geospatial object relation generation process. The reflection module also summarizes correct instances and stores them as experiential knowledge, facilitating the transfer of knowledge to subsequent tasks.

C. ORS Scene Representation

In this section, we introduce GEO-H2ACLNet, an asymmetric contrastive learning network with a hybrid architecture that employs hierarchical fusion for the joint representation of dual-temporal visual geospatial objects and structured knowledge. As illustrated in Fig. 6, GEO-H2ACLNet processes three modalities—dual-temporal ORSIs, scene graphs, and ORS scene description texts—and employs a hierarchical fusion strategy to increase the representation accuracy achieved for the relational semantics within ORS scenes through cross-modal feature interaction. GEO-H2ACLNet comprises three critical components: basic representation, image-graph feature fusion, and dual-temporal contrastive learning.

1) Basic Representation:

a) Image representation: We integrate Transformer blocks into a CNN backbone to provide relation modeling capabilities for large-scale ORS geospatial objects, thereby extracting more discriminative visual representations of the ORSI. We adopt ResNet-50 [62] as the visual encoder foundation to extract multilevel visual features. More importantly, we insert Transformer blocks into different ResNet-50 stages to capture the fine-grained geospatial object relations contained in complex ORS scenes, strengthening the representations of the geospatial object relations in multisource ORSIs. To maintain usability and accessibility, we preserve the fundamental architectures of both the CNN and attention networks. The visual encoder comprises three hierarchical feature levels: low, middle, and high levels. At the low-level, the first ResNet-50 stage extracts feature maps with sizes of C $\times$ H/4$\times$ H/4 from the given ORSI, which are then fed into the Transformer blocks to capture local dependencies, yielding local scene features encompassing individual geospatial objects. At the mid-level, the second ResNet-50 stage produces feature maps with sizes of C$\times$H/8$\times$H/8, which are then processed to explore their cross-region dependencies, generating regional features including pairwise geospatial objects and their interactive relations. At the high-level, the third and fourth ResNet-50 stages extract feature maps with sizes of C $\times$ H/32$\times$ H/32, which are then analyzed to capture global dependencies,

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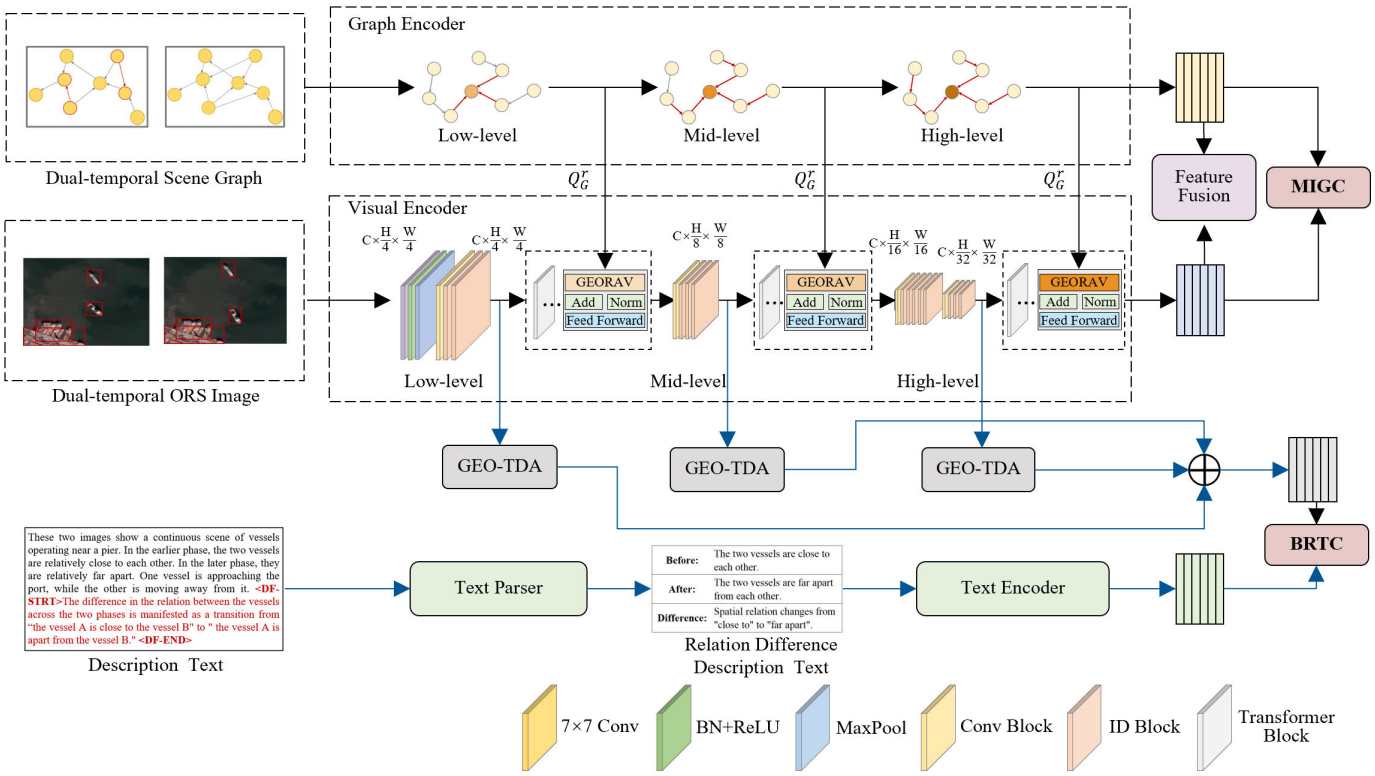


Fig. 6. Illustration of GEO-H2ACLNET.

producing global features that represent the global ORS scene semantics. We define the three-level visual feature pyramid as follows:

$$\boldsymbol{H}_{\mathrm{I},i} = \left\{H^{l}_{\mathrm{I},i}, H^{m}_{\mathrm{I},i}, H^{h}_{\mathrm{I},i}\right\} = f_{\mathrm{ImgE}}\left(X_{\mathrm{I},i}\right) \tag{8}$$

where f_{ImgE} represents the processing of the visual encoder.

b) Graph representation: Referring to the graph encoder architecture in [65], we employ a relational graph convolutional network (GCN) as a graph encoder to extract structural information from scene graphs. We encode geospatial object relational features across various subgraphs with different hop distances, constructing a multilevel graph feature pyramid. Specifically, low-level features are derived from one-hop subgraphs, including individual geospatial objects, and are aligned with the local visual features. Mid-level features are aggregated from the low-level features, encompassing the geospatial object and relation features of two-hop subgraphs and corresponding to the regional visual features. High-level features are further aggregated from the mid-level features, covering the geospatial object and relation features of three-hop subgraphs; this is consistent with global visual features. We define the three-level graph feature pyramid as follows:

$$\boldsymbol{H}_{\mathrm{G},i} = \left\{H^{l}_{\mathrm{G},i}, H^{m}_{\mathrm{G},i}, H^{h}_{\mathrm{G},i}\right\} = f_{\mathrm{GraphE}}\left(X_{\mathrm{G},i}\right) \tag{9}$$

where f_{GraphE} represents the operation of the graph encoder.

c) Text representation: To capture the relational dynamics embedded in the textual modality, we employ a Transformer-based text encoder to represent ORS scene change descriptions, which is expressed as follows:

$$H_{\mathrm{T},i} = f_{\mathrm{TextE}}\left(X_{\mathrm{T},i}\right) \tag{10}$$

where f_{TextE} represents the processing function of the text encoder.

2) Image–Graph Feature Fusion: We employ a cross-modal hierarchical fusion strategy to facilitate interactions between the visual and structural features across multiple feature levels. Based on the hybrid CNN-Transformer architecture, we introduce GEORAV-Attention to strengthen the fine-grained cross-modal interactions between ORSIs and scene graphs across three dimensions: spatial, functional, and trajectory dimensions. As shown in Fig. 7(a), we replace the multihead self-attention in the final Transformer block at each level of the visual encoder with GEORAV-Attention, achieving an optimal tradeoff between performance and the incurred computational cost. GEORAV-Attention adopts a multihead cross-attention mechanism and offers significant advantages over the original multihead self-attention mechanism. GEORAV-Attention effectively captures the intrinsic correspondences between ORSIs and scene graphs across three relation views. The learning of view-specific weights enables GEORAV-Attention to modulate cross-modal fusion outcomes, facilitating multiview feature interactions and convergence. Specifically, GEORAV-Attention takes scene graph structural features derived from three relation views as the queries Q^{SP}_{G}, Q^{SM}_{G}, and Q^{IT}_{G} and takes the visual features of the ORSIs as the keys K_I and the values V_I, expressed as follows:

$$Q^{r}_{G} = H_G W^{Q}_{G,r}, K_I = H_I W^{K}_{I}, V_I = H_I W^{V}_{I} \tag{11}$$

where $W^{Q}_{G,r}, W^{K}_{I}$, and $W^{V}_{I} \in \mathbb{R}^{d_k \times d_k}$ are learnable parameters, $r \in \boldsymbol{R} = \{SP, SM, IT\}$ represents different relation views, and d_k denotes the feature dimensionality. GEORAV-Attention

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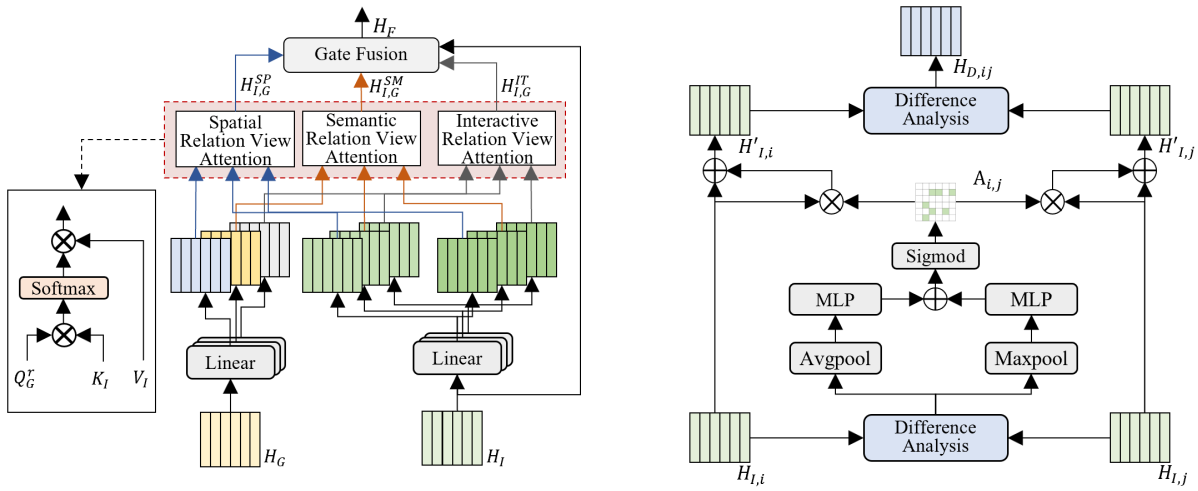


Fig. 7. Partial module interpretation of GEO-H2ACLNET. (a) GEORAV-Attention. (b) GEO-TDA.

prioritizes the most relevant content between the visual and graph features, where the attention feature can be defined as follows:

$$H^r_{I,G} = \text{softmax}\left(\frac{Q^r_G (K_I)^T}{\sqrt{d_k}}\right) V_I \tag{12}$$

where H^{SP}_I, H^{SM}_I, and H^{IT}_I denote the visual features captured from the spatial, semantic, and interactive relation views, respectively. We subsequently adopt a gated fusion module to fuse the features from acquired distinct modalities across different relation views. The fused representation is expressed as follows:

$$H_F = \left(1 - \sum_{r\in\mathbf{R}} \lambda_r\right) H_I + \sum_{r\in\mathbf{R}} \lambda_r H^r_{I,G} \tag{13}$$

$$\lambda_r = \text{gate}\left(W_I H_I + \sum_{r\in\mathbf{R}} W^r_{I,G} H^r_{I,G}\right) \tag{14}$$

where λ_r balances the contributions of the spatial, semantic, and interactive relation view attention components, and gate$(\cdot)$ denotes the gating function.

Furthermore, we perform cross-modal contrastive learning to align the ORSIs and corresponding scene graphs in a common embedding space. This process enhances the semantic consistency between the visual and structural features and improves the geospatial object relation representations across different feature levels.

3) Dual-Temporal Contrastive Learning: The geospatial object relations in ORS scenes often exhibit dual-temporal dynamics, whereas conventional image representations often emphasize holistic ORS scenes while attenuating relation change features. To explicitly capture dynamic relation evolution, we introduce GEO-TDA and perform contrastive learning between relation change texts and scene differences to strengthen the relation change representations. First, we employ a text parser [65] to extract relation change descriptions, which are subsequently processed by a text encoder to obtain H_T. Next, we input the dual-temporal ORSIs $X_{I,i}, X_{I,j}$ into an asymmetric contrastive learning network and calculate the cosine distance between the visual features $H_{I,i}, H_{I,j}$ to obtain the initial ORS scene difference $D_{i,j}$. Finally, we refine $D_{i,j}$ through GEO-TDA to produce the relation change feature H_D.

Algorithm 1 GEO-TDA Implementation Algorithm

Input: $H_{I,i}, H_{I,j}$
Output: H_D

1: $D_{i,j} \leftarrow dist(H_{I,i}, H_{I,j})$
2: $avg_pool \leftarrow AvgPool(D_{i,j})$
3: $max_pool \leftarrow MaxPool(D_{i,j})$
4: $A_{i,j} \leftarrow \sigma(MLP(avg_pool)+MLP(max_pool))$
5: $H^{attn}_{I,i} \leftarrow H_{I,i} \odot A_{i,j}$
6: $H^{attn}_{I,j} \leftarrow H_{I,j} \odot A_{i,j}$
7: $H'_{I,i} \leftarrow H_{I,i} + H^{attn}_{I,i}$
8: $H'_{I,j} \leftarrow H_{I,j} + H^{attn}_{I,j}$
9: $D'_{i,j} \leftarrow dist(H'_{I,i}, H'_{I,j})$
10: $H_D \leftarrow LinearProj(D'_{i,j})$
11: **return** H_D

As depicted in Fig. 7(b), GEO-TDA comprises four components: difference analysis, average pooling, max pooling, and an MLP. Algorithm 1 illustrates the pseudocode of the process of the proposed GEO-TDA. Following [66], we apply both average pooling and max pooling operations on $D_{i,j}$ to obtain compressed difference features (lines 1–3), which are subsequently processed through a shared-parameter MLP, and their element-wise addition yields the channel attention $A_{i,j}$ (line 4). After conducting channel-wise multiplication on the dual-temporal visual features and $A_{i,j}$ and a subsequent summation operation (lines 5–8), we perform a difference analysis again to obtain the final relation change features H_D (lines 9–10). The difference feature $D_{i,j}$ and channel attention $A_{i,j}$ are defined as follows:

$$D_{i,j} = \frac{1 - \text{dist}\left(H_{\text{I},i}, H_{\text{I},j}\right)^2}{\text{dist}\left(H_{\text{I},i}, H_{\text{I},j}\right)^2} \tag{15}$$

$$A_{i,j} = \sigma\left(\text{MLP}\left(\text{AvgP}\left(D_{i,j}\right)\right) + \text{MLP}\left(\text{MaxP}\left(D_{i,j}\right)\right)\right. \tag{16}$$

where dist$(\cdot)$ denotes the cosine feature distance, MLP denotes the multilayer perceptron, AvgP$(\cdot)$ and MaxP$(\cdot)$ denote average

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pooling and max pooling, respectively, and $\sigma(\cdot)$ denotes the sigmoid function.

We subsequently employ a linear projection layer to map the relation change features H_D and relation change text features H_T into a unified feature space for contrastive learning purposes. This enhances the discriminability of the relation change representations and alleviates the dynamic semantic ambiguity that arises from sparse fine-grained relation features.

4) Loss Function: The training process of the model is guided by a minibatch contrastive learning loss, which explicitly enhances the relational semantics and representations of relation changes through a multimodal contrastive learning strategy. The overall model contrastive loss L of the model consists of two components: MIGC loss between the ORSIs and scene graphs and DRTC loss between the dual-temporal image-scene graph pairs and relation change description text, defined as follows:

$$L = \lambda_{\mathrm{M}} L_{\mathrm{M}} + (1 - \lambda_{\mathrm{D}}) L_{\mathrm{D}} \tag{17}$$

where the MIGC loss L_{M} guides the semantic alignment process between the ORSIs and scene graphs at low, middle, and high feature levels, which is expressed as follows:

$$L_{\mathrm{M}} = \lambda_l L_{\mathrm{M}}^l + \lambda_m L_{\mathrm{M}}^m + (1 - \lambda_l - \lambda_m) L_{\mathrm{M}}^h \tag{18}$$

$$L_{\mathrm{M}}^l = -\sum_{i=1}^{N} \frac{\log \exp\left(\mathrm{sim}\left(H_{\mathrm{I},i}^l, H_{\mathrm{G},i}^l\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{sim}\left(H_{\mathrm{I},i}^l, H_{\mathrm{G},j}^l\right)\Big/\tau\right)} \tag{19}$$

where $L_{\mathrm{M}}^l, L_{\mathrm{M}}^m$, and L_{M}^h represent the contrastive losses induced at the low, middle, and high feature levels, respectively. N denotes the number of sample pairs contained in each minibatch, and $\mathrm{sim}(\cdot,\cdot)$ represents the cosine similarity function. The formulations of L_{M}^m and L_{M}^h are similar to that of L_{M}^l. The DRTC loss guides the contrastive learning process between the dual-temporal image-scene graph pairs and relation change description text, reinforcing the semantic consistency between them. This alleviates the semantic ambiguity of dynamic relations caused by sparse fine-grained relation information. The DRTC loss is formulated as follows:

$$L_{\mathrm{D}} = -\sum_{i=1}^{N} \frac{\log \exp\left(\mathrm{sim}\left(H_{\mathrm{D},i}, H_{\mathrm{T},i}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{sim}\left(H_{\mathrm{D},i}, H_{\mathrm{T},j}\right)/\tau\right)} \tag{20}$$

where $H_{D,i}$ denotes the relation change feature of sample i.

D. Multimodal Data Sampling

A high-quality dataset with rich semantics has an irreplaceable effect on method performance and training efficiency. Therefore, we construct DTITD130H, a dual-temporal image–text dataset that is used to improve the understanding of fine-grained geospatial object relations for ORSI. DTITD130H comprises 13 000 ORSIs collected from different platforms, such as EarthExplorer [67], Google Earth [68], and Esri World Imagery [69], as well as from visual tracking benchmarks such as OTB100 [70], RAIVD [71], and SatSOT [72]. More detailed information can be found in Appendix A.III (see Supplementary Material), available online.

TABLE II
DATA AND METRICS FOR ORS SGG

	Data Type	Source	Samples
Data	Image	DTITD130H	13.1k
	Text	DTITD130H	13.1k
	Relation Knowledge Text	Knowledge Database	6.1k
	Symbol	**Description**	
Metrics	Precision	Proportion of relations predicted correctly	
	Recall	Proportion of labels predicted correctly	
	F1	Harmonic mean of precision and recall	
	ICR	Invalid code rate	
	RDR	Relation diversity rate	

IV. EXPERIMENTS

To evaluate the ability of the R2C-MGR2L to comprehend the fine-grained geospatial object relations contained within ORSIs, we divide the experiments into two parts: the SGG experiment and the representation experiment. The ORS SGG experiments assess our structured representation approach in geospatial object recognition and relation modeling within complex ORS scenes. The ORS geospatial object relation representation experiments focus on the fine-grained semantic understanding of geospatial object relations across three downstream tasks: ORS relation semantic retrieval (ORSRSR), ORS relation image captioning (ORSRCP), and ORS relation-oriented visual question answering (ORSRVQA).

A. ORS SGG Experiment

1) Datasets: We conduct ORS SGG experiments based on ORSIs and multimodal knowledge extracted from DTITD130H, as summarized in Table II. These images encompass diverse, complex ORS scenes, including airports, ports, schools, transportation hubs, parks, commercial districts, residential areas, and coastal zones, providing abundant fine-grained semantics and varied geospatial object distributions. To enhance the ability of the Code-LLM for geospatial object relation recognition and reasoning, we employ RAG to supplement the inputs with three types of multimodal relation knowledge texts that are related to the images and image captions: geographic vector texts, trajectory texts, and unstructured geographic knowledge texts. Geographic vector texts are obtained from the OSM by extracting geospatial objects such as buildings, roads, and water bodies from ORSIs. Through parsing, we obtain the names, categories, and coordinates of the geospatial objects. The trajectory texts are obtained from visual tracking benchmarks such as OTB100, RAIVD, and SatSOT, which retrieve the trajectories of moving geospatial objects (e.g., airplanes, cars, and ships) that are present in the ORSIs. Unstructured geographic knowledge texts are obtained from open-source geographic knowledge databases and encyclopedic resources, which cover the definitions and functions of geospatial objects and scene descriptions.

2) Setup: We select 11 Code-LLMs trained on code corpora (Table III), which are categorized into three groups by their parameter scales: small-, medium-, and large-scale groups. To evaluate their performance in structured code generation for scene graphs, we adopt five quantitative metrics (Table II):

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TABLE III
SUMMARY OF THE RESULTS ON ORS SGG

Model	Size		Open Source	Precision	Recall	F1	ICR	RDR
CODELLAMA [75]	Small scale	7B	✓	0.17	0.13	0.15	0.85	0.17
CODEGEMMA [76]		7B	✓	0.21	0.18	0.19	0.82	0.22
GRANITE-CODE [77]		8B	✓	0.12	0.07	0.10	0.91	0.11
CODESTRAL [78]	Medium scale	22B	✓	0.51	0.41	0.45	0.60	0.31
QWEN2.5-CODER [79]		32B	✓	0.54	0.47	0.50	0.58	0.30
DEEPSEEK-CODER [80]		33B	✓	0.48	0.31	0.38	0.72	0.25
DEEPSEEK-R1-0528 [81]	Large scale	671B	✓	**0.91**	0.88	0.89	**0.01**	**0.85**
O3-2025-04-16 [82]		-	×	0.80	0.76	0.78	0.02	0.65
GMINI-2.5-PRO [83]		-	×	0.87	0.84	0.85	0.01	0.84
QWEN3-235B-A22B [84]		235B	✓	0.79	0.74	0.76	0.02	0.60
GROK-3-PREVIEW-02-24 [85]		-	×	0.65	0.59	0.62	0.02	0.50

precision, recall, the $F1$-score ($F1$), the invalid code rate (ICR), and the relation diversity rate (RDR). The ICR is defined as follows:

$$\mathrm{ICR} = \frac{N_{invalid}}{N} \tag{21}$$

where $N_{invalid}$ is the number of triplets inconsistent with the code-based ontology constraints. A higher ICR indicates lower structural stability in the observed outputs. To evaluate the semantic diversity of the results, we use the RDR, which is defined as follows:

$$\mathrm{RDR} = \frac{M_{class}}{M_{ontology}} \tag{22}$$

where M_{class} is the number of relation types in the output of each Code-LLM that conform to the code-based ontology definition, and $M_{ontology}$ is the total number of defined relation types. A higher RDR indicates richer and more diverse relational semantics.

3) Results and Analysis: We compare the performance of the tested models for Code-LLM selection. We first evaluate the small- and medium-scale Code-LLMs. As shown in Table III, these models exhibit limited performance. For example, QWEN2.5-CODER:32B achieves a precision of 0.54 but has a high ICR of 0.58, indicating substantial violations of the structural constraints. CODESTRAL:22B performs similarly poorly, with an $F1$-score of 0.45 and an ICR of 0.60, and is unable to generate reliable structured triplets. DEEPSEEK-CODER:33B and the other small-scale models perform even worse, with both their Recall and RDR values below 0.3, reflecting the restricted relation reasoning capabilities and limited semantic coverage. According to the results, DEEPSEEK-R1-0528 achieves the best performance, with a precision value of 0.91, a Recall value of 0.88, an $F1$-score of 0.89, an ICR of 0.01, and an RDR of 0.85, while generating structurally valid triplets and covering diverse, fine-grained geospatial object relations. GEMINI-2.5-PRO achieves comparable results, with an $F1$-score of 0.85 and an RDR of 0.84, when a rich set of geospatial object relations with semantically reasonable expressions is extracted. O3-2025-04-16 also performs well ($F1$ = 0.78) but suffers from semantic redundancy and misclassifications in multiobject ORS scenes. In contrast, QWEN3-235B-A22B and GROK-3-PREVIEW-02-24 often produce errors or omit geospatial objects, yielding poorer semantic expressiveness. To provide a more intuitive visual comparison, we evaluate five large-scale models, DEEPSEEK-R1-0528, O3-2025-04-16, GMINI-2.5-PRO, QWEN3-235B-A22B, and GROK-3-PRE-VIEW-02-24, as shown in Fig. 8. According to Fig. 8(b), DEEPSEEK-R1-0528 can infer complex geospatial object relations such as <Ship A, moves away, coastline> and <park, touches, coastline>. As shown in Fig. 8(c), DEEPSEEK-R1-0528 accurately captures the dynamic relation < airplane A, taxiing on, runway>. Furthermore, in Fig. 8(d), it can accurately identify the dynamic interaction between the airplanes, such as < airplane B, following, airplane A>. spanning all three types of geospatial object relations defined earlier. GEMINI-2.5-PRO achieves comparable results, with an $F1$-score of 0.85 and an RDR of 0.84, when a rich set of geospatial object relations with semantically reasonable expressions is extracted. O3-2025-04-16 also performs well ($F1$ = 0.78) but suffers from semantic redundancy and misclassifications in multiobject ORS scenes. For example, in Fig. 8(b), O3-2025-04-16 produces redundant outputs such as the spatial relation < dock structures, adjacent, water body>. In Fig. 8(c), O3-2025-04-16 misidentifies the spatial relation < taxiway, intersects, apron>. In contrast, QWEN3-235B-A22B and GROK-3-PREVIEW-02-24 often produce errors or omit geospatial objects, yielding poorer semantic expressiveness. Overall, we select DEEPSEEK-R1-0528 for subsequent experiments because of its superior balance of accuracy, semantic expression, and structural compliance.

Furthermore, we conduct comparative evaluations of SGG performance across various Code-LLM component configurations, assessing the contributions of the knowledge database, the in-context examples, and the reflective learning mechanism. We compare the performance of the tested models from three perspectives.

a) Knowledge database: To evaluate the role of domain knowledge, we conduct experiments involving each Code-LLM without relation knowledge (w/o KD). As shown in Table IV, all the models experience performance degradation to varying degrees when the multimodal relation knowledge is removed. The $F1$-score of DEEPSEEK-R1-0528 decreases from an $F1$ of 0.89–0.81, and its RDR decreases from 0.85 to 0.78, with a noticeable reduction in the number of identifiable fine-grained geospatial object relations.

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TABLE IV
COMPARISON OF RESULTS FOR ORS SGG METHODS

Model	w/o KD			Zero-shot			w/o Reflection		
	F1	ICR	RDR	F1	ICR	RDR	F1	ICR	RDR
CODELLAMA	0.13	0.85	0.16	0.14	0.89	0.17	0.15	0.86	0.16
CODEGEMMA	0.16	0.81	0.20	0.17	0.88	0.21	0.17	0.83	0.21
GRANITE-CODE	0.07	0.91	0.11	0.09	0.96	0.11	0.09	0.91	0.10
CODESTRAL	0.42	0.56	0.29	0.43	0.72	0.30	0.43	0.60	0.31
QWEN2.5-CODER	0.48	0.56	0.30	0.49	0.66	0.29	0.49	0.60	0.28
DEEPSEEK-CODER	0.36	0.67	0.24	0.37	0.80	0.22	0.38	0.73	0.24
DEEPSEEK-R1-0528	0.81	0.01	0.78	0.83	0.06	0.80	0.82	0.03	0.81
O3-2025-04-16	0.71	0.01	0.60	0.75	0.08	0.63	0.75	0.07	0.61
GMINI-2.5-PRO	0.79	0.01	0.77	0.83	0.06	0.78	0.83	0.05	0.79
QWEN3-235B-A22B	0.68	0.01	0.52	0.70	0.09	0.54	0.70	0.05	0.57
GROK-3-PREVIEW-02-24	0.54	0.02	0.42	0.58	0.11	0.44	0.59	0.05	0.48

Similarly, QWEN3-235B-A22B and GEMINI-2.5-PRO exhibit decreased performance, with their $F1$-scores decreasing by 10.5% and 7.1%, and their RDRs decreasing by 13.3% and 8.3%, respectively. In contrast, the small-scale models such as QWEN2.5-CODER:32B exhibit minimal changes, confirming their limited reasoning capabilities even when they are enhanced with domain knowledge. Overall, the introduction of domain knowledge enhances the ability of Code-LLMs to reason about geospatial object relations, with the most significant improvements observed in large-scale models.

b) In-context learning: To analyze the impact of in-context examples, we compare the performance of the tested Code-LLMs under zero-shot learning conditions. As shown in Table IV, few-shot learning generally outperforms zero-shot learning, particularly in complex, multiobject ORS scenes. For example, under zero-shot learning, DEEPSEEK-R1-0528 shows a 6.7% decrease in the $F1$-score, a 0.05 increase in the ICR, and a 5.9% decrease in the RDR, indicating a clear reduction in the structural stability and semantic richness of geospatial object relations. QWEN3-235B-A22B, despite showing good performance in few-shot learning tasks, also decreases in performance under zero-shot learning (the $F1$-score decreases by 7.9%, the ICR increases by 0.07, and the RDR decreases by 10%). Small-scale models such as CODEGEMMA:7B and CODESTRAL: 22B exhibit relatively small performance fluctuations under zero-shot learning, suggesting their inherent limitations in terms of understanding complex prompts and modeling structured geospatial object relations. Notably, GEMINI-2.5-PRO and O3-2025-04-16 demonstrate strong capabilities in SGG even under zero-shot learning ($F1$ = 0.83 and 0.75, respectively). However, few-shot learning further improves their structural stability and relation diversity. Therefore, few-shot learning facilitates the Code-LLMs in generating structurally stable triplets and reasoning about fine-grained geospatial object relations.

c) Reflective learning: To comprehensively assess the effectiveness of the reflective learning mechanism, we conduct experiments by removing the reflection module (w/o reflection). As shown in Table IV, excluding the reflection module leads to performance degradations from both the code structure compliance and the relation expression perspectives. For DEEPSEEK-R1-0528, the $F1$-score decreases by 7.9%, the RDR decreases by 4.7%, and the ICR increases from 0.01 to 0.03. GEMINI-2.5-PRO and O3-2025-04-16 also significantly decrease, with their $F1$-scores decreasing by 2.4% and 3.8%, respectively, and their ICRs increasing by an average of 0.04, resulting in more incorrect geospatial object relations. In contrast, smaller scale models such as QWEN2.5-CODER:32B and CODESTRAL: 22B exhibit minor fluctuations (their $F1$ and RDR values decrease by approximately 0.01) without reflective learning, highlighting their inherently limited ability to reason about geospatial object relations and the limited benefits derived from reflective learning. Overall, reflective learning, through a closed-loop optimization process involving error recognition, strategy feedback, and experience transfer, enhances the accuracy and semantic expression capabilities of larger scale models in geospatial object relation reasoning scenarios.

B. ORS Representation Experiments

To substantiate the effectiveness of our proposed framework in terms of performing geospatial object relation representation for ORS scenes, we conduct experiments on three relation-aware downstream tasks: ORSRSR, ORSRCP, and ORSRVQA. In accordance with the results of the SGG experiments, DEEPSEEK -R1-0528 is selected as the Code-LLM for generating scene graphs and training GEO-H2ACLNet. In addition, the experimental datasets for these downstream tasks are constructed using multimodal knowledge derived from geographic vector data, trajectory data, and geographic knowledge texts.

1) Datasets: As shown in Table IV, the datasets for each downstream task are composed of two parts: mainstream datasets and self-developed datasets. The mainstream datasets primarily consist of ORSIs captured by platforms such as WorldView-2/-3, GeoEye-1, and Sentinel-2. The self-developed datasets, which are derived from DTITD130H, comprise three task-specific subsets—DT-RSR, DT-RCP, and DT-RVQA—that are designed to evaluate the relation representation capabilities of the proposed method. Moreover, DTITD130H is further partitioned into a training subset and an inference subset, with the latter held out entirely from the training process and thus remaining unseen to both the compared methods and our own model. Under this setting, all methods are evaluated under consistent conditions, eliminating any potential advantage arising from data familiarity.

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TABLE V
DATA AND METRICS FOR ORS REPRESENTATION EXPERIMENT

ORSRSR	Data	Dataset	Samples	Dataset Processing
		RSICD [86]	18.2k	Geospatial object relation enrichment
		DT-RSR	2.4k	Data derivation from DTITD130H
	Metrics	Symbol	Description	
		RA-R@k	Relation-aware recall at top-k	
		RA-mAP	Relation-aware mean average precision	
ORSRCP	Data	Dataset	Samples	Dataset Processing
		RSICD	18.2k	Geospatial object relation enrichment
		DT-RCP	2.4k	Data derivation from DTITD130H
	Metrics	Symbol	Description	
		RA-BLEU	Relation-aware bilingual evaluation understudy	
		RA-METEOR	Relation-aware metric for evaluation of translation with explicit ordering	
		RA-CIDEr	Relation-aware consensus-based image description evaluation	
		RA-ROUGE_L	Relation-aware longest common subsequence	
ORSRVQA	Data	Dataset	Samples	Dataset Processing
		RSIEval [48]	0.9k	Complex sample filtering and Q&A augmentation
		EarthVQA [87]	6k	Complex sample filtering and Q&A augmentation
		VRSBench [89]	123k	Complex sample filtering and Q&A augmentation
		DT-RVQA	8.7k	Data derivation from DTITD130H
	Metrics	Symbol	Description	
		RA-OA	Relation-aware overall accuracy	
		RA-AA	Relation-aware average accuracy	

2) Setup: In our experiments, a GCN [65] is utilized as a graph encoder for the structural representation of scene graphs. For visual representation purposes, ResNet-50-V1.5 [64] is adopted as the backbone of the visual encoder within GEO-H2ACLNet to extract multiscale visual features. Transformer blocks are inserted after the first, second, and fourth stages to achieve multilevel cross-attention. Following the practices in [90], the low and middle levels incorporate three Transformer blocks to avoid excessive modification of local and regional features, whereas the high-level incorporates six Transformer blocks to facilitate global feature integration. For text representation, the CLIP-ViT B/32 encoder [91], comprising 12 Transformer blocks with balanced scale and representation performance, is employed in GEO-H2ACLNet to extract textual features for scene relation descriptions. Linear projection layers are applied to harmonize the dimensional consistency across multimodal encoder outputs. Between the first-stage Code-LLM and the second-stage GEO-H2ACLNet, we introduce an external processing module that leverages expert-defined rules and professional remote sensing tools to filter and validate the generated knowledge triplets. Triplets exhibiting substantial deviations are removed, thereby enhancing the accuracy of the extracted relational triplets. We employ DeepSpeed [93] to train GEO-H2ACLNet, leveraging model and data parallelism for distributed training across multiple GPUs and nodes. During training, we adopt AdamW [94] as the optimizer for its decoupled weight decay regularization. As shown in Table VI initial learning rate is set to $\eta_0 = 0.0001$ with a linear warm-up phase spanning the first 10 epochs, [95] followed by a cosine annealing schedule defined as follows:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})\left(1 + \cos\left(\frac{t - T_w}{T - T_w}\pi\right)\right) \tag{23}$$

where $\eta_{\min}$ denotes the minimum learning rate, T_w denotes the warm-up epochs, and T denotes the total training epochs.

To ensure a fair comparison with the intrinsic capabilities of each baseline (BL) method, we leverage the few-shot in-context learning capability of compared methods, such as general VLMs (Qwen3, Gemini-2.5) and RS-specific models (EarthVQA, RS-LLaVA) for moderate adaptation, avoiding the additional cost introduced by model training.

As shown in Table V, we adopt two metrics for the performance evaluation: the relation-aware recall at k (RA-R@k) and the relation-aware mean average precision (RA-mAP). RA-R@k denotes the proportion of correct results among the top-k retrieval results and is defined as follows:

$$\text{RA} - \text{R@k} = \frac{\text{TP@}k}{\text{TP@}k + \text{FN@}k} \tag{24}$$

where TP@k represents the number of correct results, and FN@k denotes the number of incorrect results. Specifically, RA-R@1, RA-R@5, and RA-R@10 correspond to the recalls calculated from the top 1, 5, and 10 retrieval results, respectively. RA-mAP is an extension of the conventional mAP metric, where mAP is calculated for each relation type separately to examine the comprehensive performance achieved by methods across different relation views. The RA-mAP is defined as follows:

$$\text{RA} - \text{mAP} = \sum_{r\epsilon\{SP,SM,IT\}} \lambda_r \sum_{n=1}^{N_r} \text{mAP} \tag{25}$$

where N_r denotes the number of samples of relation type r, $r\epsilon\{SP, SM, IT\}$.

For the ORSRCP task, we adopt four enhanced evaluation metrics for performance evaluation purposes: RA-BLEU4 (RA-B4), RA-METEOR (RA-M), RA-CIDEr (RA-C), and RA-ROUGE_L (RA-R). These metrics extend the conventional text generation evaluation measures by incorporating a weighting mechanism introduced for relation keywords. RA-BLEU is adapted from the conventional BLEU metric, which measures the similarity between generated and reference texts by dividing them into n-grams and applying a sentence length difference penalty. In our adaptation, the

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length calculation exclusively retains relational semantic content while irrelevant information is filtered, thereby reducing the evaluation bias. RA-BLEU is defined as follows:

$$\text{RA}-\text{BLEU} = BP \times \exp\left(\sum_{n=1}^{N} W_n \times \log P_n\right) \tag{26}$$

where P_n is the precision of the n-grams, and W_n is the weight of each n-gram. BP is a penalty factor concerning text length differences, which can be expressed as follows:

$$BP = \begin{cases} 1, & l_c > l_s \\ \exp\left(1 - \dfrac{l_r}{l_c}\right), & l_c \le l_s \end{cases} \tag{27}$$

where l_c denotes the length of the generated description, and l_s is the length of the reference text that is closest to l_c. RA−METEOR is modified from the conventional METEOR metric, which measures the performance of a method based on the weighted harmonic mean of its unigram precision and recall while incorporating an additional penalty term, expressed as follows:

$$\text{RA}-\text{METEOR} = (1 - Pen) \times F_{mean} \tag{28}$$

where Pen represents the penalty, and F_{mean} is the harmonic mean of the precision and recall. RA−$CIDEr$ is derived from the standard $CIDEr$ metric, which measures the lexical diversity and consistency between generated text and multiple reference texts and is formulated as follows:

$$\text{RA}-CIDEr(c_i, S_i) = \sum_{n=1}^{N} \omega_n CIDEr_n(c_i, S_i) \tag{29}$$

$$CIDEr_n(c_i, S_i) = \frac{1}{M} \sum_{j=1}^{M} \frac{g^n(c_i) \times g^n(s_{ij})}{\|g^n(c_i)\| \times \|g^n(s_{ij})\|} \tag{30}$$

$$g_k(c_i) = \lambda_k TF(k) \times IDF(k) \tag{31}$$

where c_i denotes the description of the ith ORSI, s_{ij} is the jth reference text associated with image i, M is the number of reference texts per image, k denotes the kth n-gram, and λ_k denotes the assigned weight reflecting n-gram alignment with target relational patterns. RA−ROUGE_L is an enhanced version of the ROUGE_L metric to improve the evaluation process by focusing on the continuity and consistency of relation phrases through longest common subsequence (LCS) matching, expressed as follows:

$$R_{lcs} = \frac{LCS(X, Y)}{m} \tag{32}$$

$$P_{lcs} = \frac{LCS(X, Y)}{n} \tag{33}$$

$$\text{RA}-\text{ROUGE}_\text{L} = F_{lcs} = \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \tag{34}$$

where R_{lcs} and P_{lcs} denote the recall and precision, respectively, and $LCS(X, Y)$ is the length of the longest relational-aware common subsequence between X and Y, and $\beta = P_{lcs}/R_{lcs}$.

For the ORSRVQA task, we employ the relation-aware average accuracy (RA-AA) and the relation-aware overall accuracy (RA-OA) as evaluation metrics. These metrics extend the conventional AA and OA measurements and evaluate the model's performance in terms of its ability to recognize visual entities and geospatial object relations by comparing the generated results with ground-truth answers. Values approaching 1.0 indicate superior performance.

3) Results and Analysis:

a) Orsrsr results: The ORSRSR task involves retrieving ORSIs that satisfy the geospatial object relations described in natural language queries. We employ a relation-aware k-nearest neighbor (R-KNN) strategy to retrieve the top-K images that are most consistent with the described relations. Comparisons are conducted among R2C-MGR2L, PIR-CLIP [45], RemoteCLIP [46], and SWPE [88].

Table VII presents the results on the RSICD and DT-RSR datasets, which demonstrate that the performance of the R2C-MGR2L is superior in terms of both the RA-R@k and the RA-mAP. With respect to the RSICD, R2C-MGR2L outperforms PIR-CLIP by 8.11% and 10.96% in terms of the RA-R@10 and the RA-mAP, respectively. With respect to DT-RSR, R2C-MGR2L demonstrates substantial improvements, surpassing PIR-CLIP by 18.91% for RA-R@10% and 22.76% for RA-mAP. Fig. 9 provides visual comparisons, where correct results are highlighted in green, incorrect results are shown in red, and partially reasonable but unannotated results are presented in yellow. As shown in Fig. 9, SWPE and RemoteCLIP retrieve images that are consistent with global scene features but fail to capture fine-grained relations (e.g., misclassifying a truck as a bus or misidentifying an elevated road as a circular square), limiting their ability to model relational semantics. Moreover, these comparison methods struggle to capture geospatial object relations in complex ORS scenes (e.g., recognizing that an airplane is located south of an airport). In contrast, the R2C-MGR2L demonstrates distinct advantages in terms of modeling spatial and interactive relations, yielding more accurate retrievals.

We carry out extensive ablation experiments to investigate the individual contribution of each constituent in the R2C-

TABLE VI
PARAMETER SETTINGS OF THE R2C-MGR2L

Categories	Parameters	Value
Model Training Parameters	Optimizer	AdamW
	Minibatch size	100
	Epoch	200
	Learning rate	0.0001
MIGC Loss Weights	multimodal loss weight λ_M	0.73
	relation change loss weight λ_D	0.17
DRTC Loss Weights	low-level feature loss weight λ_l	0.2
	mid-level feature loss weight λ_m	0.1

TABLE VII
COMPARISON OF R2C-MGR2L WITH SOTA ORSRSR METHODS

Methods	RSICD				DT-RSR			
	RA-R@1	RA-R@5	RA-R@10	RA-mAP	RA-R@1	RA-R@5	RA-R@10	RA-mAP
SWPE	0.153	0.392	0.535	0.393	0.103	0.346	0.463	0.352
RemoteCLIP	0.145	0.395	0.558	0.372	0.126	0.375	0.502	0.351
PIR-CLIP	0.291	0.534	0.678	0.511	0.232	0.514	0.645	0.492
ours	**0.314**	**0.592**	**0.733**	**0.567**	**0.358**	**0.631**	**0.767**	**0.604**

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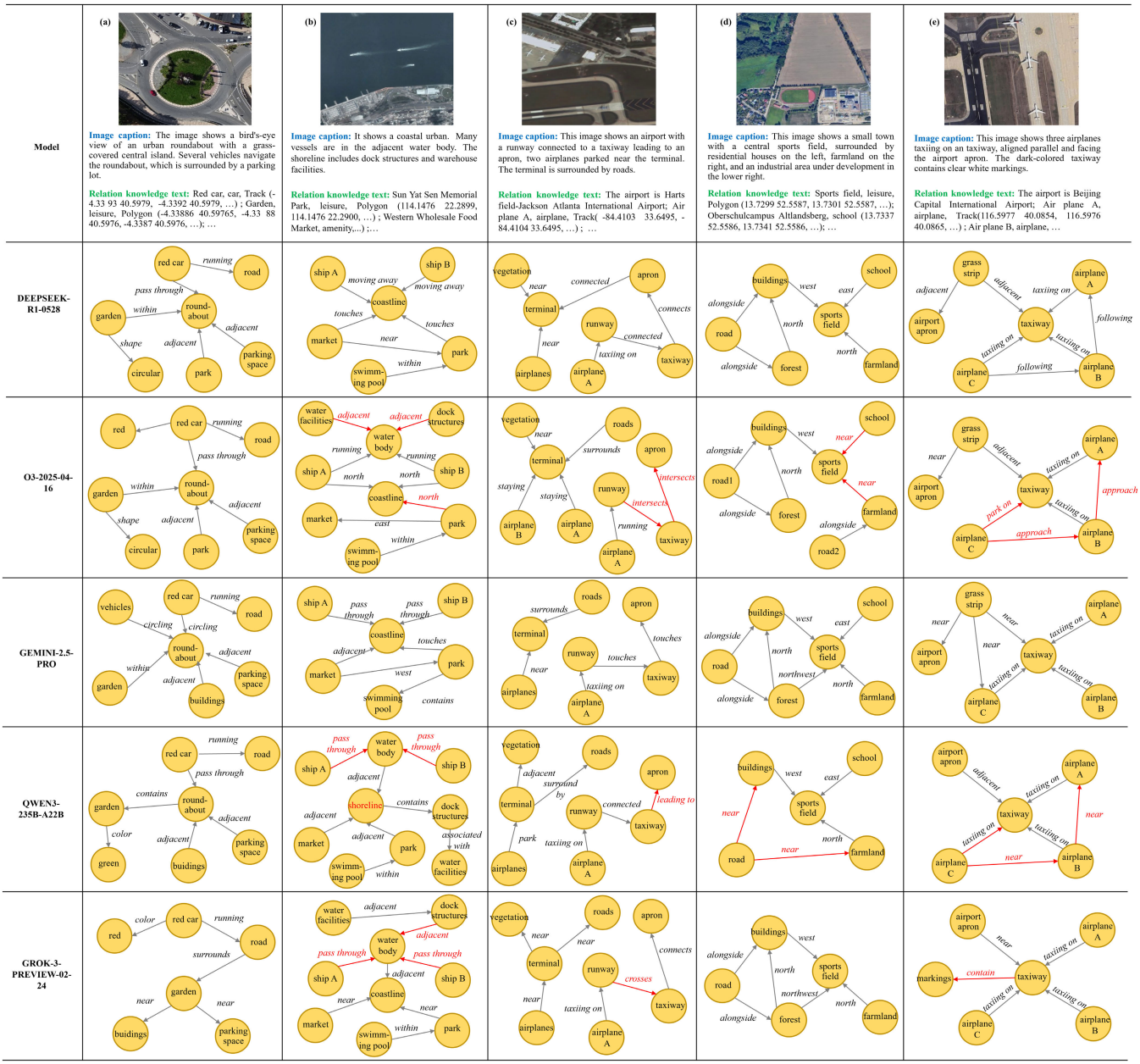


Fig. 8. Visualization results of ORS scene graphs generated by five large-scale Code-LLMs.

TABLE VIII
RESULTS OF ABLATION MODEL STUDIES ON ORSRSR

Ablation Architecture	RSICD				DT-RSR			
	RA-R@1	RA-R@5	RA-R@10	RA-mAP	RA-R@1	RA-R@5	RA-R@10	RA-mAP
BL + SMI	0.254	0.513	0.669	0.487	0.272	0.564	0.692	0.546
BL + MMI	0.273	0.533	0.695	0.512	0.334	0.603	0.73	0.580
BL + MMI + GEO-H2ACLNet	0.297	0.536	0.697	0.518	0.34	0.613	0.741	0.591
BL + MMI + GEO-H2ACLNet + GEORAV-Attention	0.304	0.548	0.704	0.537	0.354	0.638	0.752	0.594
BL + MMI + GEO-H2ACLNet + GEORAV-Attention + GEO-TDA	0.312	0.589	0.731	0.576	0.357	0.643	0.763	0.606

MGR2L method. With the basic image encoder as the BL, we compare the performance of the R2C-MGR2L across different component combinations. For clarity, we provide abbreviations for the ablated modules: single-modal image input (SMI), multimodal image-graph input (MMI), GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA. The results in Table VIII indicate that the MMI, GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA modules contribute

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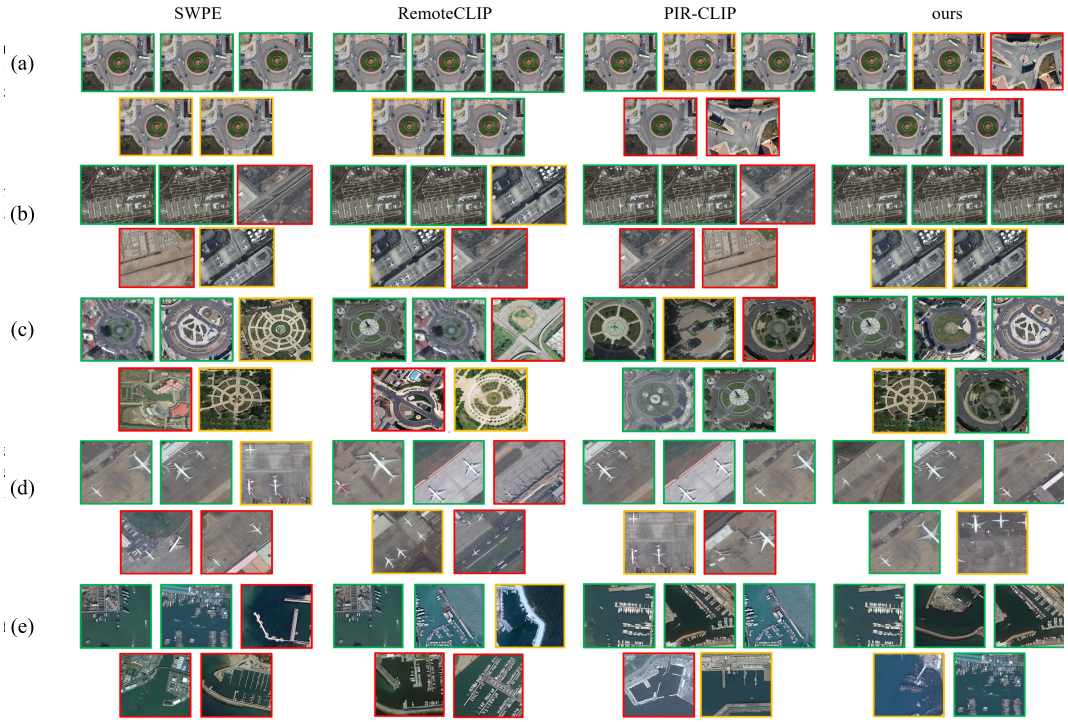


Fig. 9. Visualization comparison of R2C-MGR2L with SOTA ORSRSR. Green, red, and yellow highlight correct, incorrect, and partially reasonable but unannotated results, respectively. (a) Dark-blue care travels counter-clockwise within the southern area of the roundabout. (b) On the southern apron, aircraft are neatly arranged, ready for operations. (c) Circular square which is surrounded by roads. (d) Two aircraft are parked parallel to each other at the airport, in the same direction. (e) Ships are navigating near a port.

TABLE IX
ATTENTION COMPARISON RESULTS ON ORSRSR

Attention	RSICD				DT-RSR			
	RA-R@1	RA-R@5	RA-R@10	RA-mAP	RA-R@1	RA-R@5	RA-R@10	RA-mAP
ORG-Attention	0.289	0.523	0.682	0.493	0.316	0.575	0.696	0.549
CRS-Attention	0.295	0.531	0.685	0.497	0.322	0.581	0.702	0.557
GEORAV-Attention (SP)	0.299	0.539	0.699	0.508	0.347	0.618	0.737	0.585
GEORAV-Attention (SM)	0.296	0.53	0.693	0.519	0.323	0.602	0.723	0.552
GEORAV-Attention (IT)	0.298	0.556	0.718	0.532	0.345	0.625	0.744	0.577
GEORAV-Attention (SP+SM+IT)	0.309	0.587	0.734	0.569	0.353	0.641	0.759	0.605

significantly to the performance of the R2C-MGR2L, with consistent improvements achieved in all four metrics for the ORSRSR task. The complete architecture (BL + MMI + GEO-H2ACLNet + GEORAV-Attention + GEO-TDA) achieves the best performance, with a 12.85% improvement over the BL.

To verify the impact of GEORAV-Attention, we conduct further comparisons across different attention mechanisms: original attention (ORG-Attention), cross-modal attention (CRS-Attention), and geospatial object relation-aware view attention (GEO-RAV-Attention) under various configurations. The results in Table IX indicate that both cross-modal attention and relation-aware attention contribute to the performance of the model in the ORSRSR task. When GEORAV-Attention incorporates all the relation views (SP, SM, and IT), R2C-MGR2L achieves the best performance, with its RA-mAP improving by 11.01% over that of ORG-Attention. These findings confirm that multiview modal interaction enables complex geospatial object relations to be more comprehensively modeled, leading to improved retrieval accuracy.

b) Orsrcp results: The ORSRCP task involves generating descriptive texts for ORSIs, covering visual geospatial objects and their relations. Comparisons are conducted with three state-of-the-art methods: MG-Transformer [47], RSGPT [48], and RS-LLava [49]. Table X reports the results obtained on the RSICD and DT-RCP. The results show that the R2C-MGR2L method consistently outperforms all the comparison methods. With respect to the RSICD, compared with RS-LLava, the best BL, R2C-MGR2L, improves the RA-B4, RA-C, and RA-R metrics by 27.65%, 14.04%, and 17.84%, respectively. With respect to DT-RCP, R2C-MGR2L achieves even greater improvements. Fig. 10 presents qualitative examples, with spatial relations, semantic relations, and interactive relations in outputs highlighted in blue, yellow, and green, respectively. The MG-Transformer captures global semantics but exhibits biases in fine-grained recognition tasks. For instance, the model fails to accurately identify vehicles on the road and their spatial relations, as shown in Fig. 10(a). RSGPT and RS-LLaVA achieve slightly better geospatial

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Fig. 10. Visualization comparison of proposed R2C-MGR2L with SOTA ORSRCP methods in different ORS scenes. (a) Traffic roundabout, (b) industrial, (c) traffic roundabout, (d) waterway transport, (e) airport, (f) industrial area, (g) river, (h) airport. GT represents Ground Truth. In the results, blue markers indicate spatial relations, yellow markers denote semantic relations, and green markers identify interactive relations. Within the given ORSIs, regions associated with these relations are visualized using colors that align with the corresponding ground-truth labels.

object recognition but remain limited in geospatial object relation modeling capabilities, particularly for orientation and topological relations. The RSGPT inaccurately identifies the geographic orientation of land features and mislabels rivers

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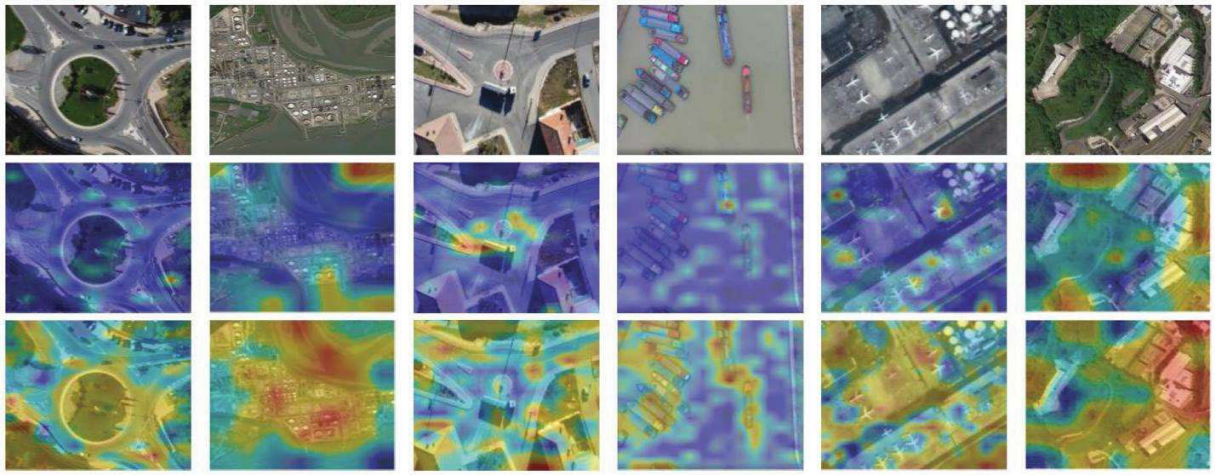


Fig. 11. Partial visualization of the heatmap of attention comparison on ORSRCP. The first row depicts the given ORSIs. The second row depicts the ORG-Attention heatmap. The third row depicts the proposed GEORAV-Attention heatmap. The color gradient represents the attention distribution, with warmer colors (red) indicating higher attention to spatial regions and cooler colors (blue) reflecting lower attention.

TABLE X
COMPARISON OF R2C-MGR2L WITH SOTA ORSRCP METHODS

Dataset	Methods	RA-B4	RA-M	RA-R	RA-C
RSICD	MG-Transformer	0.462	0.343	0.618	2.847
	RSGPT	0.368	0.309	0.523	1.031
	RS-LLava	0.528	**0.607**	0.648	2.36
	ours	**0.674**	0.576	**0.739**	**2.781**
DT-RCP	MG-Transformer	0.398	0.298	0.588	2.013
	RSGPT	0.629	0.344	0.724	2.239
	RS-LLava	0.688	0.592	0.618	2.331
	ours	**0.825**	**0.654**	**0.910**	**3.005**

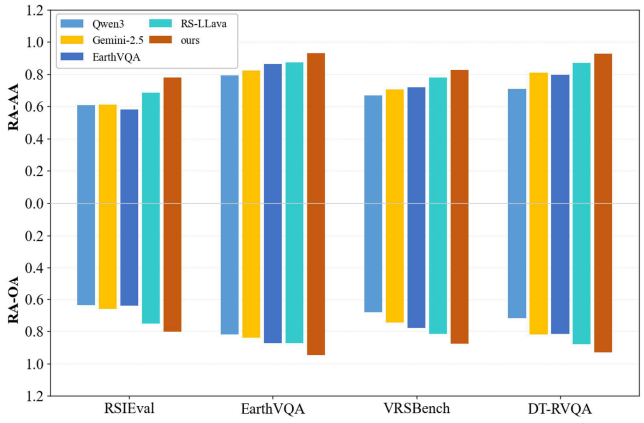


Fig. 12. Comparison of R2C-MGR2L with various state-of-the-art ORSRVQA methods in terms of RA-AA and RA-OA.

and wetlands distributed from the northeast to the northwest. RS-LLaVA erroneously places oil tanks near the developed area, as shown in Fig. 10(b). In contrast, the R2C-MGR2L demonstrates superior performance. In Fig. 10(a), the R2C-MGR2L correctly identifies the vehicle moving southwest along the exit. Thus, the R2C-MGR2L provides more accurate descriptions with notable spatial and interactive relation modeling improvements.

Table XI presents the ablation results produced across different configurations, combining the BL with SMI, MMI, GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA. The results show that the performance improves consistently as the modules are integrated. Specifically, GEO-H2ACLNet contributes the most substantial gains, whereas GEORAV-Attention and GEO-TDA provide further enhancements. The complete R2C-MGR2L method achieves RA-R improvements of 9.36% and 21.61% over the BL + SMI on the RSICD and DT-RCP, respectively. These findings indicate that MMI, GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA play critical roles in enhancing the ORSRCP performance of the model.

Table XII compares the performances achieved by the R2C-MGR2L across different attention mechanisms: ORG-Attention, CRS-Attention, and GEORAV-Attention. The results show that ORG-Attention, which considers only intramodal dependencies, fails to capture cross-modal information, yielding suboptimal results. GEORAV-Attention significantly boosts performance, and when integrating all relation views (SP, SM, and IT), R2C-MGR2L achieves the optimal results, with the RA-C attained on the RSICD reaching 2.781, a 24.32% improvement over ORG-Attention. As shown in Fig. 11, ORG-Attention focuses primarily on a few geospatial objects while ignoring others, treating many geospatial objects and their relations as background information. Conversely, GEORAV-Attention captures multiple geospatial objects and their complex relations, enabling richer and more accurate spatial, semantic, and interactive descriptions and ultimately yielding the best performance.

c) Orsrvqa results: The ORSRVQA task involves answering natural language questions about geospatial object relations in ORSIs through semantic understanding and logical reasoning. Unlike relation captioning, ORSRVQA emphasizes local content that is specifically relevant to questions involving ORS scenes. Comparisons are conducted with two general-purpose vision-language models, i.e., Qwen3 [84] and Gemini-2.5 [83], and two domain-specific models, i.e., EarthVQA [87] and RS-LLava [49]. The comparative results

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TABLE XI
RESULTS OF ABLATION MODEL STUDIES ON ORSRCP

Ablation Architecture	RSICD				DT-RCP			
	RA-B4	RA-M	RA-R	RA-C	RA-B4	RA-M	RA-R	RA-C
BL + SMI	0.612	0.531	0.673	2.136	0.701	0.562	0.745	2.078
BL + MMI	0.638	0.548	0.693	2.23	0.725	0.579	0.796	2.566
BL + MMI + GEO-H2ACLNet	0.643	0.551	0.711	2.445	0.73	0.582	0.822	2.716
BL + MMI + GEO-H2ACLNet + GEORAV-Attention	0.664	0.572	0.702	2.652	0.756	0.603	0.834	2.901
BL + MMI + GEO-H2ACLNet + GEORAV-Attention + GEO-TDA	0.674	0.573	0.736	2.774	0.825	0.654	0.906	3.003

obtained are presented in Fig. 12. R2C-MGR2L outperforms EarthVQA and RS-LLava by 7.44% and 6.76% in RA-AA, respectively. This superiority becomes more pronounced on DT-RVQA and other datasets. The results of the qualitative comparisons are shown in Fig. 13, with correct and incorrect outputs highlighted in green and red, respectively. While Qwen3 and Gemini-2.5 exhibit strong reasoning capabilities for directional judgments (e.g., distinguishing clockwise from counterclockwise vehicle motion), their analyses remain confined to image contents and lack grounding in geospatial semantics. Consequently, these methods struggle with spatial orientation and absolute positional reasoning [e.g., determining the geographic directions of vehicles or aircraft, in Q2 of Fig. 13(a)]. They also suffer from hallucinations, such as when predicting nonexistent geospatial objects [e.g., pedestrians or ships in Q1; Fig. 13(f)]. The comparative methods further fail to capture the relations highlighted in the questions. For example, in Q1 of Fig. 13(d), most methods identify the vehicles on the main loop road but overlook the relation queries regarding the vehicles traveling alongside the white bus. In addition, errors frequently occur in reasoning tasks concerning dynamic interactions because of insufficient domain-specific knowledge (e.g., misclassifying a moving train as stationary). Through multimodal knowledge integration and joint image-scene graph representation, the R2C-MGR2L accurately models fine-grained spatial topologies, relative positions, and dynamic interactions. This enables the R2C-MGR2L to capture question-relevant content, achieving superior performance.

The ablation results obtained across different configurations when the BL is combined with the SMI, MMI, GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA on the RSIEval, EarthVQA, VRSBench, and DT-RVQA datasets are shown in Fig. 14(a). The results show that integrating MMI, GEO-H2ACLNet, GEORAV-Attention, and GEO-TDA substantially improves the performance of R2C-MGR2L. Notably, GEO-TDA provides significant contributions, improving the RA-AA values by 13.94% and 14.70% on EarthVQA and DT-RVQA, respectively. The results produced by the proposed method with different attention mechanisms, namely, ORG-Attention, CRS-Attention, and GEORAV-Attention, are compared in Fig. 14(b). These results demonstrate that GEORAV-Attention yields substantial gains, improving the RA-AA by 12.61% on EarthVQA and by 13.42% on DT-RVQA. This finding indicates that performing cross-modal fusion across multiple relation views enables the R2C-MGR2L to achieve more comprehensive and accurate relation understanding.

Fig. 15 provides the attention heatmaps produced by ORG-Attention and GEORAV-Attention for intuitive illustration purposes. ORG-Attention predominantly focuses on salient features (e.g., traffic–island flowerbeds) while neglecting other geospatial objects and their relations (e.g., vehicles on loop roads). In contrast, GEORAV-Attention focuses on multiple geospatial objects and their complex relations, achieving superior performance with enhanced accuracy.

d) Hyperparameter experiments: In addition, we conduct hyperparameter experiments to determine the optimal parameter settings for the R2C-MGR2L by gradually increasing the parameter values within the [1] interval. Two parameter groups are explored: the multimodal loss weight λ_{M} and relation change loss weight λ_{D}, as well as the multilevel feature loss weights λ_l and λ_m. Specifically, λ_{M} and λ_{D} balance the contributions of L_{M} and L_{D}, while λ_l and λ_m balance the contributions of the low-, mid-, and high-level contrastive losses L_{M}^{l}, L_{M}^{m} and L_{M}^{h}, respectively.

The RA-mAP values achieved by the R2C-MGR2L in the RSRSR task under different combinations of λ_{M} and λ_{D} are shown in Fig. 16(a). As observed in Fig. 16(a), when $\lambda_{\mathrm{M}} = 1$ and $\lambda_{\mathrm{D}} = 0$, only the multimodal contrastive loss is considered, yielding an RA-mAP of 0.638. When λ_{D} increases to approximately 0.17, the RA-mAP reaches 0.656, at which point the R2C-MGR2L achieves optimal outcomes. However, further increases in λ_{D} result in accuracy degradation. These findings confirm that multimodal relational semantics govern the effectiveness of R2C-MGR2L and that the integration of a small number of relation change features remains essential for achieving optimal performance.

Fig. 16(b) illustrates the impact of varying λ_l and λ_m on the RSRSR accuracy of the model. When $\lambda_l = 0$ and $\lambda_m = 0$, only high-level features are considered, and the RA-mAP remains approximately 0.589. As λ_l and λ_m increase, the performance of the R2C-MGR2L gradually improves. At $\lambda_l = 0.2$ and $\lambda_m = 0.1$, the RA-mAP reaches approximately 0.661, representing the optimal configuration of the R2C-MGR2L. Further increases in λ_l and λ_m lead to a decline in accuracy. The results confirm that while high-level relational semantics are crucial for the relation representation ability of the R2C-MGR2L, low- and mid-level features provide valuable additional performance gains.

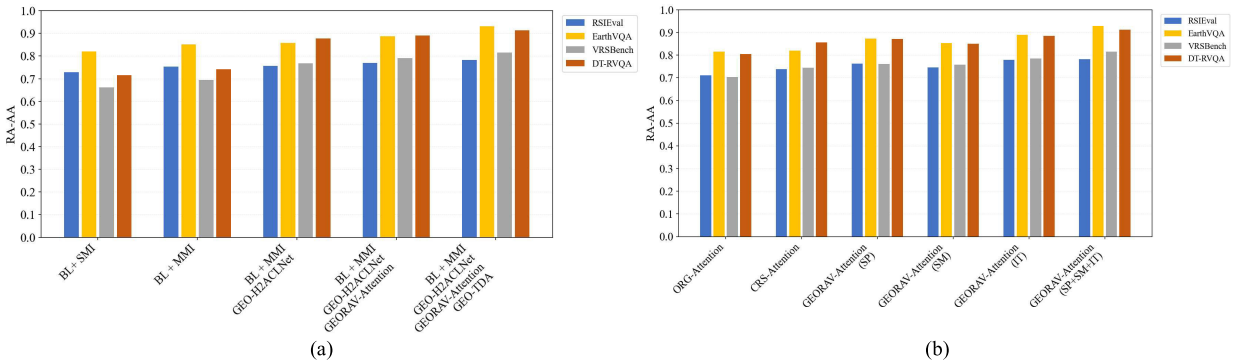
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Fig. 13. Visualization comparison of R2C-MGR2L with SOTA ORSRVQA methods in different ORS scenes. (a) Airport, (b) park, (c) traffic roundabout, (d) port, (e) railway, (f) traffic roundabout, (g) port, (h) river, (i) highway, (j) highway, (k) bridge, (l) airport, (m) highway, and (n) railway. In the results, green markers indicate correct content, red markers denote erroneous content. Yellow markers in ORSIs indicate the regions associated with the geospatial object relations mentioned in each question.

e) Error propagation experiments: SGG errors can propagate in a cascading manner. Inaccurate scene graph triplets introduce biased relational representations into the hierarchical feature fusion process, which degrades the quality of the fused multimodal features and ultimately impairs downstream task performance. We conduct an error propagation experiment to quantify the relationship between first-stage SGG quality and downstream task performance and evaluate R2C-MGR2L under different combinations of error correction measures. In the experiment, we apply ablation settings to three error correction modules: the RE and the reflection mechanism (RM) in the first stage, and the external processing (ED)

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(a) (b)

Fig. 14. Results of ablation studies and attention comparison on ORSRVQA. Different colors represent the RA-AA values across different datasets. (a) Ablation studies on model modules. (b) Attention comparison.

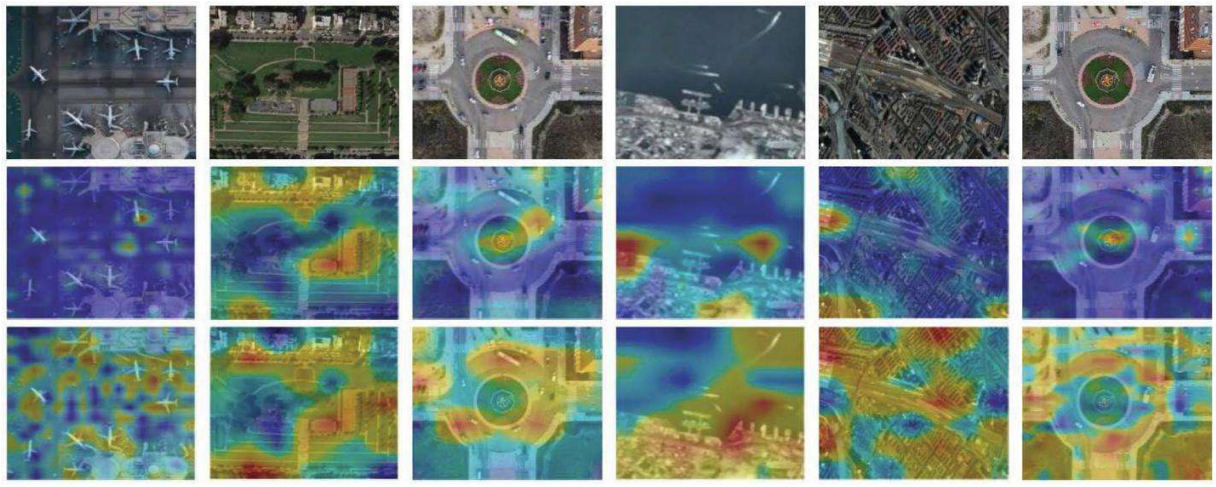
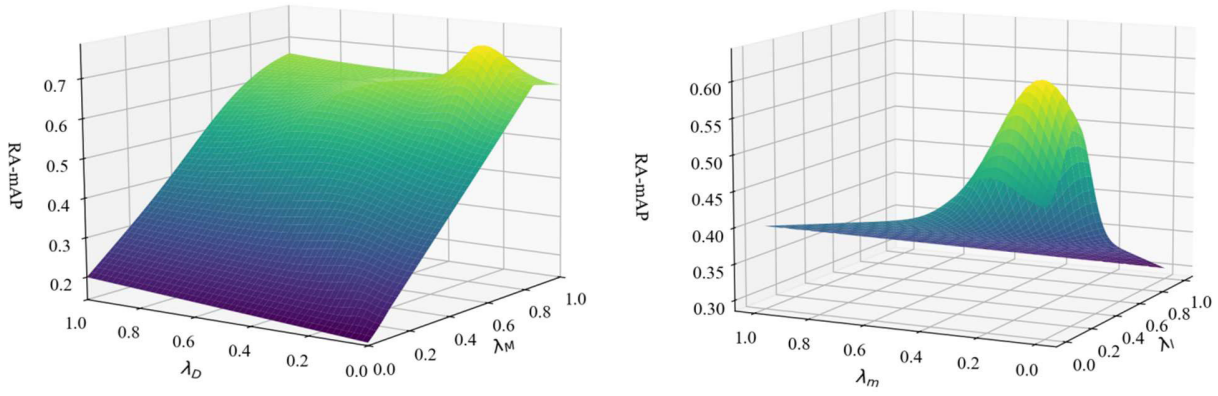


Fig. 15. Partial visualization of the heatmap of attention comparison on ORSRVQA. The first row displays the given ORSIs. The second row presents the ORG-Attention heatmap. The third row displays the proposed GEORAV-Attention heatmap. The color gradient represents the attention distribution, with warmer colors (red) indicating higher attention to spatial regions and cooler colors (blue) reflecting lower attention.



(a) (b)

Fig. 16. RA-mAP results from the loss function parameter experiment conducted on the ORSRSR task with the DT-RSR dataset. (a) Impact of hyperparameter λ_M and λ_D. (b) Impact of hyperparameter λ_l and λ_m.

module. The results are presented in Table XIII. Analysis reveals that as the first-stage SGG $F1$ increases from 0.76 to 0.89, all downstream task metrics exhibit consistent improvement, indicating that the first-stage SGG generation quality exerts a significant positive propagation effect on downstream task performance. Meanwhile, different downstream tasks exhibit varying sensitivity to errors: ORSRSR demonstrates relatively high robustness, as its embedding representations capture holistic scene semantics with inherent tolerance to local relation errors; ORSRCP is most sensitive to errors, since caption generation depends more directly on the correctness of specific relational descriptions; and ORSRVQA exhibits inter-

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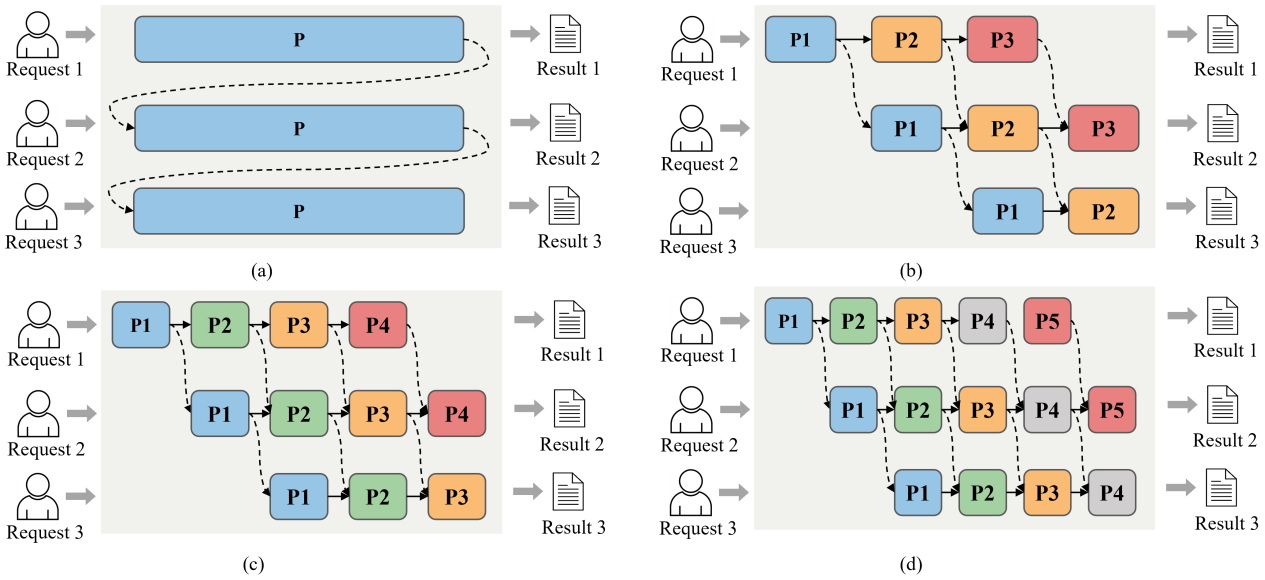


Fig. 17. Illustration of different execution strategies. Solid arrows indicate data flow directions, and dashed arrows indicate control execution order. (a) Illustration of SE mode. (b) Illustration of 3PPE mode. (c) Illustration of 4PPE mode. (d) Illustration of the 5PPE mode.

TABLE XII
ATTENTION COMPARISON RESULTS ON ORSRCP

Attention	RSICD				DT-RCP			
	RA-B4	RA-M	RA-R	RA-C	RA-B4	RA-M	RA-R	RA-C
ORG-Attention	0.617	0.513	0.671	2.237	0.703	0.566	0.728	2.152
CRS-Attention	0.633	0.549	0.687	2.649	0.729	0.585	0.802	2.783
GEORAV-Attention (SP)	0.652	0.559	0.715	2.69	0.742	0.594	0.816	2.824
GEORAV-Attention (SM)	0.627	0.553	0.704	2.663	0.717	0.589	0.807	2.795
GEORAV-Attention (IT)	0.646	0.556	0.726	2.678	0.735	0.591	0.812	2.868
GEORAV-Attention (SP+SM+IT)	0.678	0.569	0.734	2.781	0.821	0.665	0.907	3.012

TABLE XIII
RESULTS OF ERROR PROPAGATION ON DOWNSTREAM PERFORMANCE

ORS SGG	ABALATION			ORSRSR				ORSRCP				ORSRVQA	
F1	RE	RM	ED	RA-R@1	RA-R@5	RA-R@10	RA-mAP	RA-B4	RA-M	RA-R	RA-C	RA-AA	RA-OA
0.76	√			0.305	0.589	0.708	0.548	0.714	0.576	0.825	2.436	0.812	0.763
0.76	√	√		0.310	0.595	0.714	0.554	0.722	0.583	0.832	2.478	0.818	0.769
0.76	√	√	√	0.317	0.603	0.721	0.562	0.735	0.593	0.842	2.548	0.826	0.778
0.78	√			0.321	0.608	0.726	0.567	0.742	0.598	0.847	2.584	0.831	0.782
0.78	√	√		0.328	0.615	0.733	0.574	0.756	0.608	0.856	2.642	0.839	0.790
0.78	√	√	√	0.339	0.626	0.745	0.587	0.778	0.625	0.874	2.756	0.852	0.805
0.85	√			0.348	0.635	0.754	0.596	0.798	0.637	0.889	2.871	0.862	0.814
0.85	√	√		0.351	0.638	0.757	0.600	0.806	0.642	0.894	2.912	0.867	0.819
0.85	√	√	√	0.354	0.641	0.761	0.604	0.818	0.650	0.901	2.968	0.872	0.824
0.89	√			0.357	0.643	0.763	0.606	0.825	0.654	0.906	3.003	0.876	0.827
0.89	√	√		0.362	0.649	0.769	0.612	0.838	0.663	0.914	3.058	0.883	0.834
0.89	√	√	√	0.371	0.658	0.778	0.621	0.856	0.678	0.926	3.145	0.893	0.845

mediate sensitivity with moderate fluctuations. Furthermore, under each SGG $F1$ level, the progressive addition of the RE, RM, and external processing modules yields stable incremental performance gains. Among them, the ED module—serving as an external processing module embedded with expert rules and professional remote sensing tools—effectively suppresses error propagation to downstream tasks, providing reliable assurance for downstream task performance. In summary, the experimental results demonstrate that our multilayered error correction mechanisms effectively intercept the propagation of errors from the SGG stage to downstream tasks.

f) Computational complexity experiment: We conduct a computational complexity experiment to quantify the efficiency and practical applicability of R2C-MGR2L. Our proposed R2C-MGR2L is a two-stage framework that mitigates efficiency concerns through decoupled processing. Building upon this design, we introduce a four-phase parallel execution (4PPE) strategy from an engineering optimization perspective, decomposing the inference process into four phases: multimodal knowledge retrieval (P1), ORS SG construction (P2), ORS scene representation (P3), and downstream task execution (P4). We adopt a pipelined scheduling strategy

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TABLE XIV
COMPUTATIONAL COMPLEXITY OF MODULES IN R2C-MGR2L

Processes of R2C-MGR2L	Module	Params (M)	FLOPs (G)
P1: Multimodal Knowledge Retrieval	RAG Module	22	0.5
P2: ORS SG Construction	Code-LLM	671,000	—
	Reflection Module	0.1	0.01
P3: ORS Scene Representation	Visual Encoder	25.6	4.1
	Graph Encoder	52.8	8.6
	Text Encoder	4.2	0.8
	GEO-TDA Module	87.7	4.4
	Linear Projection Layers	2.1	0.3
P4: Downstream Task Execution	Downstream Task Heads	3.8	0.5

TABLE XV
TOTAL PROCESSING TIME UNDER VARYING NUMBERS OF REQUESTS

Request Number	SE Mode	3PPE Mode		4PPE Mode		5PPE Mode	
	Total Time (s)	Total Time(s)	Reduction (%)	Total Time(s)	Reduction (%)	Total Time(s)	Reduction (%)
100	119.7	81.4	32.0	57.8	51.7%	66.8	44.2
500	611.5	403.6	34.0	259.5	55.1%	316.5	48.2
1000	968.8	598.7	38.2	412.5	57.4%	483.6	50.1
5000	4836.5	2952.3	38.9	2054.8	57.5%	2418.3	50.0
10000	9681.2	5898.5	39.1	4115.3	57.5%	4841.2	50.0

to overlap different requests across phases—e.g., when the first request is at P4, the second request simultaneously executes P3, and the third proceeds with P2. Furthermore, within P1, each request is decomposed into multiple retrieval conditions, enabling concurrent multimodal knowledge retrieval to further improve efficiency.

We evaluate sequential execution (SE) and three parallel execution strategies with different phase granularities—three-phase parallel execution (3PPE), four-phase parallel execution (4PPE), and five-phase parallel execution (5PPE)—under varying numbers of requests (100, 500, 1000, 5000, and 10 000) to comprehensively assess scalability (see Fig. 17). In the SE mode, all processing steps for each request are executed sequentially, with subsequent requests waiting until the current request finishes entirely. In the 3PPE mode, the inference process is decomposed into three phases: multimodal knowledge retrieval and ORS SG construction (P1), ORS scene representation (P2), and downstream task execution (P3). In the 4PPE mode, the inference process is decomposed into four phases: multimodal knowledge retrieval (P1), ORS SG construction (P2), ORS scene representation (P3), and downstream task execution (P4). In the 5PPE mode, the inference process is further decomposed into five phases: multimodal knowledge retrieval (P1), ORS SG construction (P2), basic representation (P3), image–graph feature fusion (P4), and downstream task execution (P5). Table XIV reports the parameter counts and FLOPs for each module, and Table XV presents the total processing time under SE and the three parallel execution strategies across varying numbers of requests. According to the results in Table XV, IIIPPE provides moderate time reductions over SE, achieving 38.2% at 1000 requests, as the merged first phase creates a bottleneck that limits pipelining efficiency. 4PPE demonstrates the best performance, with time reductions of 51.7% at 100 requests, 55.1% at 500 requests, and 57.4% at 1000 requests. Notably, at 5000 requests, the time reduction stabilizes at 57.5% and remains at 57.5% at 10 000 requests, confirming that the pipeline reaches saturation at approximately 5000 requests and the optimization approaches its practical upper bound. However, 5PPE yields notably diminished returns compared to 4PPE, with time reductions consistently lower across all request volumes, as the excessive decomposition introduces substantial interphase synchronization overhead that outweighs the benefits of finer-grained pipelining. These results confirm that 4PPE achieves the optimal balance between phase granularity and pipelining efficiency, and effectively addresses the computational overhead introduced by Code-LLM inference while maintaining practical scalability for real-world deployment scenarios with concurrent geospatial analysis demands.

V. CONCLUSION

This work addresses the limitations of the prevailing methods in terms of recognizing and reasoning about multiscale geospatial object relations in complex ORS scenes. We propose a framework that enhances multigranularity relational representation in dual-temporal ORS images through LLM-based SGG for ORSI. Our approach adopts a two-stage learning paradigm to facilitate a comprehensive understanding of various geospatial object relations. In the first stage, we leverage a pretrained Code-LLM to perform programmatic modeling of three relation types among multiscale geospatial objects and generate corresponding knowledge triplets, thereby attaining enhanced relational semantic modeling. In the second stage, we introduce an asymmetric contrastive learning network with a hybrid architecture for the geospatial object representation. This network performs hierarchical fusion for

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joint representation of dual-temporal visual geospatial objects and structured knowledge across different feature depths. Experiments conducted on relation-aware downstream tasks demonstrate that our framework outperforms the state-of-the-art methods and achieves superior accuracy, completeness, and timeliness in relational semantic recognition scenarios. Our framework can effectively support high-precision geospatial object detection and relation recognition tasks for the intelligent interpretation of complex ORS scenes.

Despite the promising performance, several limitations warrant further discussion. While the two-stage modular design allows the scene graphs generated in stage one to be cached and reused for multiple downstream queries, the quality of SGG is inherently bounded by the reasoning capability of the Code-LLM, and the initial generation cost remains a practical concern for time-critical applications, where a complete end-to-end solution has not yet been achieved. Moreover, the current framework is designed for bi-temporal analysis and requires further extension to accommodate tri-temporal or multitemporal sequences for long-term continuous change monitoring. In future work, we plan to encapsulate the two-stage pipeline into a unified workflow service to improve overall efficiency and support real-time demands. We also intend to explore additional relation dimensions, such as relation representation under external environmental disturbances and motion interference in low-altitude observation scenarios, to further enrich the scene graph representation.

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